

The Value of Shade: Canopy Loss and Mortality in Urban Mexico

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Abstract

We estimate mortality effects of urban canopy loss, instrumenting with nearby forest insect outbreaks. Where outbreaks measurably reduce canopy and remaining shade is limited, loss raises mortality by 6.3 percent during hot, dry, sunny April–June but has no detectable effect during the also-hot, cloudier July–September. Effects concentrate in cardiovascular mortality, span working-age and older adults, and appear regressive by insurance status. Converging evidence points to shade as the primary channel. When heat and solar exposure coincide, losing this shared service implies about 120 additional deaths per year. Conserving existing canopy is thus a valuable component of urban heat adaptation.

JEL Codes: Q54, Q57, I14, I15, I18

Keywords: urban canopy, heat mortality, ecosystem services, climate adaptation, health inequality, Mexico

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1 Introduction

Air conditioning has substantially reduced heat mortality where it is widely available ([Barreca et al., 2016](#)), but it protects people only while they occupy cooled spaces. Urban canopy can provide a shared complement by shading people, buildings, and surfaces. Urban forest policy often involves incremental decisions about conserving existing canopy, yet cities have little evidence on the mortality cost of canopy loss at this margin, when that cost arises, or who bears it. Identifying the causal effect of canopy loss is difficult because development and other local changes can affect both canopy and health.

We estimate the causal effect of losing existing urban canopy on mortality at this margin and use the pattern of effects to assess whether shade is the relevant protective service. Mexico is well suited to this question. Heat exposure is intense and seasonal; air-conditioning adoption is growing rapidly but from a low base ([Davis and Gertler, 2015, 2025](#)); and a large informal workforce cannot always avoid outdoor exposure or poorly cooled workplaces. Weather-related mortality falls disproportionately on poorer households in Mexico, a gradient established for cold days but not yet resolved for heat ([Cohen and Dechezleprêtre, 2022](#)). This makes distributional incidence a central question. The setting also offers an unusual combination of data: satellite measures of canopy loss, ground-surveyed forest-health records, and the universe of registered deaths. We assemble these records for an analysis sample of 4,110 urban localities observed from 2002 to 2023. These units are census-delineated footprints of inhabited urban settlements. We measure canopy loss within those footprints and mortality among their residents. Together, the records allow us to examine seasonal timing, cause-specific mortality, and distributional incidence.

We instrument canopy loss with outbreaks of tree-damaging insects recorded in nearby forest plots outside locality boundaries. This is a plausible instrument because nearby outbreaks predict canopy loss within urban localities, while these insects are not known to harm human health. Nonetheless, environmental conditions could jointly affect outbreaks and mortality, and outbreaks could influence mortality through channels other than urban canopy loss. Our central falsification examines low-cover localities, where outbreaks remain present but do not measurably change urban canopy. A mortality response there would cast doubt on a canopy-loss interpretation. We complement this test with flexible controls for environmental conditions and falsifications based on other ecological disturbances. We report two complementary estimates. A Poisson control function gives the model-based mortality effect at the measured canopy-loss margin. The reduced form directly estimates the outbreak-induced mortality contrast without dividing by the first-stage coefficient.

The modest scale of the identifying variation is intrinsic to the policy question. In the

full sample, evaluated at the mean locality area, a nearby outbreak increases measured annual canopy loss by about 0.13 hectares, a small annual change relevant to decisions over existing trees and canopy patches rather than wholesale canopy removal. The design therefore identifies the mortality cost of canopy loss at a scale relevant to urban forest conservation. This limited physical footprint makes shade a natural candidate channel and motivates a test of whether locality-wide ambient cooling can explain the mortality response. Section 3.2 develops four empirical implications of that channel. First, mortality should not respond where outbreaks do not reduce canopy. Where they do, effects should tend to be larger where less substitute shade remains. Second, effects should concentrate in hot periods with high solar exposure and weaken under cloud cover. Third, they should concentrate in heat-sensitive causes of death. Fourth, they should be larger among heat-vulnerable groups and those with fewer private means of avoiding exposure.

The results broadly follow these predictions. In localities with very little baseline canopy, outbreaks do not measurably increase canopy loss, and mortality does not respond either. Among localities where outbreaks do increase canopy loss, the mortality point estimate is larger at medium than at high baseline cover, consistent with fewer substitutes for lost shade at medium cover, although the two estimates are not statistically distinguishable. In the medium-cover sample, the instrument-induced canopy-loss shift raises all-cause mortality in April–June, the hot, dry, sunny quarter, by 6.3% under the control function ($p = 0.001$). The corresponding reduced form is 6.2% ($p < 0.001$). The estimate remains positive at 4.5% when medium- and high-cover localities are pooled ($p = 0.001$). The 6.3% medium-cover estimate corresponds to roughly one additional April–June death per outbreak-exposed locality-year, or about 120 additional deaths annually across the localities in that sample. Applying a Mexico-adjusted value of a statistical life gives an average annual mortality cost of approximately \$339 million over the 2002–2023 study period.

Seasonal timing, cloud cover, and locality-wide temperature all reinforce the shade interpretation. The mortality estimate is largest in April–June and statistically larger than in every other quarter. The most informative seasonal comparison is with July–September, which is also hot but substantially cloudier: the April–June estimate exceeds it by 5.7 percentage points ($p < 0.001$). Within April–June, the estimate is small in unusually overcast years but positive and statistically significant under normal and unusually sunny conditions, although the contrast is imprecisely estimated ($p = 0.066$). We then ask whether the instrument generates enough locality-wide heating to explain the mortality response. The estimated shift in locality-wide air temperature is, if anything, in the opposite direction, and even under conservative assumptions thermal changes at that scale can account for at most 6.5% of the April–June response.

Cause-of-death patterns provide complementary evidence consistent with heat stress: cardiovascular mortality rises by 7.3% ($p = 0.001$), while we detect no clear response in respiratory or external-cause mortality, although both estimates are imprecise. The incidence results are consistent with multiple dimensions of heat vulnerability. Mortality rises among adults aged 65 and older (+6.1%, $p = 0.002$), consistent with physiological susceptibility, and among working-age adults (+4.5%, $p = 0.024$), but not among children. Within working age, the largest point estimate occurs among those outside formal employment-based insurance (+8.8%, $p = 0.035$), consistent with greater exposure or fewer means of avoiding and managing heat.

We make two contributions. First, we extend research on urban trees by identifying and valuing the mortality consequences of losing established canopy at an incremental conservation margin. Evidence from studies of pest-driven tree loss and planting programs links urban trees to higher property values and lower energy use (Han et al., Forthcoming), improved infant health (Jones and Goodkind, 2019), and lower mortality (Donovan et al., 2013, 2022). The mortality studies, however, leave unresolved which protective services generate the response. Other work traces pollution-removal (Tan, 2022) and pollen (Xing et al., 2026) pathways, while broad-area assessments infer health benefits from changes in ambient temperature (Jones, 2019; Iungman et al., 2023). Taken together, our evidence on where and when mortality rises, which causes respond, and who bears the burden points to shade as the main protective service. Established canopy can therefore provide consequential health protection through shade even when that protection does not register in locality-average temperature.

Second, we contribute to the economics of climate adaptation by estimating the mortality value of existing urban canopy and tracing the incidence of its loss in a middle-income setting where private cooling is limited and unevenly distributed. Existing work shows that temperature-related mortality varies with income, climate, and adaptive capacity (Heutel et al., 2021; Carleton et al., 2022) and that households reduce heat risk through costly private cooling (Deschênes and Greenstone, 2011). That protection is tied to cooled settings: in Indian manufacturing, workplace cooling eliminated on-site productivity losses but not heat-related absenteeism (Somanathan et al., 2021). More broadly, Carleton et al. (2024) distinguish ex ante investments from ex post responses to realized weather, identify public goods and equity as rationales for adaptation policy, and call for evidence on when the benefits of specific interventions exceed their costs. Canopy conservation is such an ex ante margin. It keeps shade in place before hot weather arrives and can complement private cooling by protecting shared outdoor spaces and uncooled buildings without requiring individual action at the moment of exposure. Our estimates provide the benefit side of that assessment

under prevailing avoidance and private adaptation. That value is state contingent, arising when heat and solar exposure coincide. The distribution of this mortality burden also matters in Mexico, where prior evidence finds that heat mortality falls disproportionately on younger adults and points to occupational exposure as a plausible channel (Wilson et al., 2024; Bressler et al., 2026). We extend this evidence by showing that working-age mortality rises when a specific source of heat protection is lost, with a larger response among those outside formal social-insurance coverage, an imperfect proxy for employment conditions and adaptive capacity. Together, these patterns suggest unequal incidence and bear on the equity rationale for adaptation policy.

2 Data

Our analysis links satellite measures of tree cover and loss to ground-based forest-health surveys, individual death records, population data, and gridded climate and environmental measures. This section describes how we construct each locality-level measure and assemble the analysis sample. Appendix Table A1 lists the source, spatial resolution, and coverage of each dataset.

2.1 Data Sources and Measurement

We measure baseline canopy and annual loss using the Hansen Global Forest Change dataset, derived from Landsat imagery at 30-meter resolution (Hansen et al., 2013). We aggregate the pixels within the 2010 boundaries of urban localities defined by Mexico’s National Institute of Statistics and Geography (INEGI) (INEGI, 2010). Baseline canopy cover is the percentage of locality area occupied by pixels with at least 1 percent tree cover in 2000. Annual canopy loss is the percentage of locality area assigned a loss year in each year from 2001 through 2023. The annual measure is therefore a flow of newly detected loss, not the stock of canopy remaining in that year.

Hansen records loss when a stand-replacing disturbance is detected within a 30-meter pixel and assigns that loss to a calendar year. It therefore captures substantial tree mortality or removal but may miss partial crown dieback, thinning, small street trees, and other fine-scale losses of shade. The measure does not identify whether the loss was natural or anthropogenic, and its annual timing does not establish whether a recorded loss occurred before or after mortality observed within the same year.

As a measurement check, we also use the NASA Carbon Monitoring System (CMS)

continuous tree-cover product for Mexico, available at 30-meter resolution from 2000 to 2019 (Tran et al., 2025). Unlike Hansen’s discrete loss indicator, CMS measures tree cover continuously within each pixel and can therefore register partial changes that do not trigger Hansen’s loss classification. Taking annual differences in this continuous estimate produces a noisier change measure, however, and the series covers a shorter period, so we use it only as an alternative canopy measure in Appendix Table B3. Because both products derive from Landsat reflectance, the comparison tests sensitivity to the canopy-measurement algorithm rather than to the underlying satellite sensor.

Data on insect outbreaks come from Mexico’s National Forest and Soil Inventory (INFyS), administered by the National Forestry Commission (CONAFOR) (CONAFOR, 2004–2020). The inventory provides tree-level observations from systematically sampled forest plots (conglomerados) across Mexico’s forested areas, covering three survey rounds (2004–2009, 2009–2014, 2015–2020). Plot locations follow a systematic grid stratified by ecological and geographic criteria, independently of forest health conditions or insect presence.

For each plot-year, we calculate the plot-level insect-damage share as the fraction of sampled trees exhibiting damage attributed to defoliators, bark beetles, or wood-boring insects. We link these observations to urban localities by calculating the distance from each locality boundary to every INFyS plot within 15 km, the largest radius considered in our sensitivity analysis. We interpret nearby plots as a sample of insect damage in the surrounding forest, not as point sources from which particular insects must travel to urban trees. Because INFyS samples forested land, nearby plots generally lie outside locality boundaries. Only about 1% of urban-locality polygons contain an INFyS plot (52 polygons), and reconstructing the preferred instrument after excluding all inside-boundary plots changes no locality-year classifications. The canopy-loss measure itself remains restricted to the area inside the urban locality boundary.

We convert the plot observations into a binary instrument using four parameters: a distance radius, a within-plot damage threshold, a required share of qualifying plots, and a carry-forward window. The preferred instrument uses plots within 10 km of the locality boundary. A surveyed plot-year qualifies when at least 20% of its sampled trees exhibit insect damage. A locality-year records a qualifying detection when at least 30% of the nearby plots surveyed that year qualify. A year with no nearby plot surveyed cannot generate a new qualifying detection. The instrument equals one in the recording year and the following four years. Each later qualifying detection begins another five-year window, so overlapping windows extend the period during which the instrument remains active.

The spatial and damage parameters serve distinct purposes. Across all urban localities, approximately 93% of locality-years have no surveyed plot within 5 km, leaving that

radius too sparse. A 15 km radius draws on more distant forest conditions and, at the preferred damage thresholds and window, produces a weaker first stage. We therefore use 10 km to balance survey coverage and spatial proximity. The 20% within-plot threshold lies above the median insect-damage share among plot-years with any recorded damage, approximately 13%, and therefore selects relatively severe damage. The 30% requirement uses the prevalence of severe damage among the available nearby observations rather than a simple any-damage indicator.

The five-year carry-forward window reflects both the survey process and the timing of observed canopy loss. INFyS revisits each plot roughly once every five years, so a qualifying detection dates a scheduled recording visit rather than biological onset. Tree mortality may also reach a coarse satellite measure with delay: after bark-beetle attack, foliage can remain visually unchanged for months, while reddening and needle loss unfold over subsequent years (Wulder et al., 2006; Meddens et al., 2013), and repeated defoliation can cause mortality over several years (De Grandpré et al., 2019). Consistent with these lags, an event-level analysis shows cumulative canopy loss after the recording visit rising through the first three to four years and then plateauing (Appendix B.2.2). This pattern supports carrying the recorded detection forward for five years, but it does not identify the duration of the biological outbreak because onset remains interval-censored. Appendix Figure B2 evaluates sensitivity to all four construction choices.

For falsification tests, we construct parallel indicators of mistletoe and fire damage using the same spatial and temporal parameters. Mistletoe captures nearby biological stress to trees. Infection spreads slowly and kills hosts over many years rather than in a pulse (Hawksworth and Wiens, 1996), so it is unlikely to generate the abrupt canopy loss that annual satellite data record. Fire captures nearby non-insect damage to surveyed trees and tests whether predictive power for canopy loss is specific to insect damage rather than ecological disturbance more generally.

Mortality data come from Mexico’s vital-statistics system, which contains the universe of registered deaths from 2002 to 2023 (INEGI, 2023). Each record identifies the decedent’s locality of residence, month and year of death, age, sex, cause of death, occupation, and health-insurance affiliation. We assign deaths to locality of residence and year of occurrence rather than year of registration, and aggregate them into annual and quarterly locality-level counts. Quarters follow the calendar year.

We classify causes of death as cardiovascular (ICD-10 chapter I), respiratory (chapter J), or external (chapters V–Y, including accidents, homicides, and suicides). Age groups are 0–14, 15–64, and 65 and older. We classify IMSS, ISSSTE, military, PEMEX, and private insurance as formal coverage, and the uninsured and Seguro Popular as informal

coverage. Insurance affiliation is used as a proxy of broader exposure and vulnerability, including working conditions or access to cooling. We also harmonize reported occupations into primary-sector, sales and services, technical or manual, and professional categories. The first two form the broader outdoor classification and the latter two the indoor classification.

Population counts come from Mexico’s population and housing censuses of 2000, 2010, and 2020 and the 2005 population count (INEGI, 2021). We construct annual locality populations between census years by applying municipal growth rates from the National Population Council’s annual projections to the locality census benchmarks (CONAPO, 2024). Quarterly population exposure equals one-fourth of the annual total. Cause-specific outcomes use total population, and age-specific outcomes use the corresponding age-group population. Insurance and occupation outcomes use the working-age population because annual denominators for those subgroups are unavailable.

We assemble quarterly climate and environmental measures for 2001–2023 to characterize seasonal and year-to-year conditions, control for potential climate and smoke confounding, and bound thermal changes visible at the locality scale. MODIS cloud frequency and ERA5-Land surface solar radiation characterize cloud conditions and surface solar exposure; the radiation measure includes both direct and diffuse components. Cloud frequency is the quarterly share of daily 1-km MOD09GA observations at the locality centroid classified as cloudy or mixed. ERA5-HEAT provides the Universal Thermal Climate Index and mean radiant temperature. ERA5-Land provides daily mean and maximum 2-m air temperature, 2-m dewpoint temperature, and total precipitation; we derive relative humidity from the daily mean air and dewpoint temperatures before averaging it within each quarter. Temperature-day bins use daily average temperature defined as $(T_{\max} + T_{\min})/2$, following the heat-mortality literature (Barreca et al., 2016; Cohen and Dechezleprêtre, 2022). TerraClimate supplies the Palmer Drought Severity Index. MODIS daytime land-surface temperature, averaged across each locality polygon, provides a higher-resolution thermal measure. MODIS aerosol optical depth at the locality centroid and active-fire detection-days within 10 km measure ambient smoke and nearby burning. These satellite smoke measures are distinct from the CONAFOR fire-damage indicator used in the falsification test, which records damage to trees in surveyed forest plots. We also assign each locality to its 1991–2020 Köppen-Geiger climate zone for the climate-zone-by-year robustness specification.

To describe how baseline canopy varies with deprivation across urban localities, we use CONEVAL’s year-2000 Social Lag Index, a multidimensional measure combining educational attainment, access to health services, housing quality, basic dwelling services, and household assets.

2.2 Urban Localities and Analysis Samples

Our unit of analysis is the urban locality as delineated in Mexico’s 2010 census. INEGI classifies a settlement as urban if it has at least 2,500 residents or serves as a municipal or state seat. The census universe comprises 4,458 urban localities. We link mortality, population, and Social Lag records using INEGI’s nine-digit locality identifier. Raster and gridded measures are assigned using the locality polygons, centroids, or buffers described above, while INFyS plots are linked by distance to the locality boundary. We assemble mortality, population, canopy, and the outbreak indicator at the locality-year level and attach environmental measures by locality, year, and quarter.

The balanced mortality panel requires Hansen-detected baseline canopy in 2000, leaving 4,149 localities. It spans 2002–2023 and contains 91,278 locality-year observations across 31 of Mexico’s 32 states. Baja California is the only state not represented because none of its urban localities has detectable baseline canopy in the Hansen data. For the annual all-cause estimation, we exclude four locality-years with zero population; fixed-effect separation leaves 90,416 observations across 4,110 localities. Quarterly and subgroup samples vary slightly because separation is outcome-specific.

The Social Lag analysis instead uses a separate cross-sectional match to the 4,458-locality census universe. It contains 4,197 localities: 3,969 with Hansen-detected baseline canopy and 228 with none. This cross-section and the balanced mortality panel are not nested: the Social Lag cross-section includes those 228 zero-canopy localities, while 180 positive-canopy localities in the balanced mortality panel lack a Social Lag match.

Localities in the annual all-cause estimation sample have approximately 19,960 residents and cover 5.15 square kilometers on average. Roughly one-quarter are smaller settlements classified as urban because they are administrative seats. Their census polygons delineate compact, inhabited settlements rather than surrounding rural areas. On average, 88.8% of the area within these boundaries lies within 100 meters of grid cells containing buildings in the Global Human Settlement Layer. The canopy measure therefore includes trees that shade the places where people live, work, and walk, although it does not distinguish them from other trees within the settlement. Figure 1 displays the spatial distinction at the center of the design: canopy and recorded canopy loss are measured inside the settlement, whereas the nearby forest-inventory plots used to construct the outbreak instrument lie outside its boundary but within 10 kilometers.

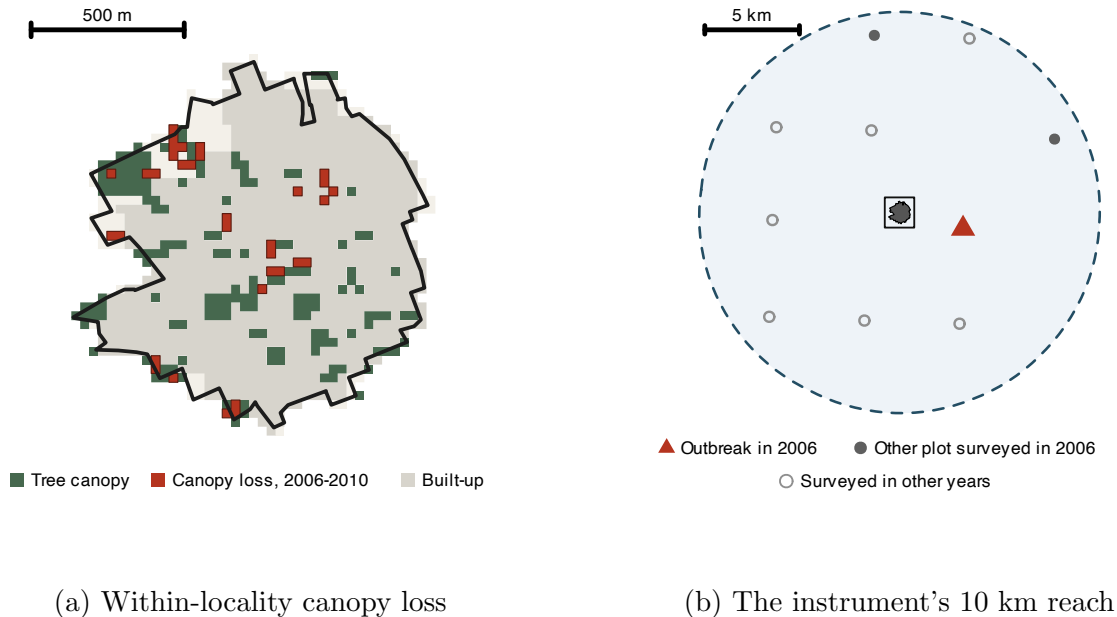


Figure 1: Urban Canopy Loss and Nearby Insect Outbreaks

Notes: Plan de Ayala, Chiapas, is an illustrative urban locality with a nearby insect outbreak recorded in 2006. In Panel (a), red denotes canopy loss during 2006–2010, the five-year instrument window beginning in the recording year; green denotes tree canopy, and grey denotes built-up surface in 2000. Canopy and loss are from Hansen Global Forest Change; built-up surface is from the Global Human Settlement Layer. Panel (b) shows the 10 km search buffer measured from the locality boundary; the square marks the area shown in Panel (a). The triangle marks the outbreak plot, filled circles mark other forest-inventory plots surveyed in 2006, and hollow circles mark plots surveyed only in other years.

3 Empirical Strategy

3.1 Identification

Canopy loss is endogenous because localities that lose canopy may differ systematically from those that do not, and development pressures that remove trees may independently affect health. We therefore instrument locality-level canopy loss with nearby forest insect outbreaks. Identification comes from changes over time in a locality's exposure to these outbreaks, with locality fixed effects absorbing permanent differences across places and year fixed effects absorbing common annual shocks. Beyond instrument relevance, a causal interpretation requires that outbreaks affect mortality only through the canopy loss they induce (*exclusion*) and that within-locality changes in outbreak exposure be unrelated to other time-varying determinants of mortality (*independence*).

We begin with exclusion. The instrument is constructed from insect damage recorded in forest plots outside the boundaries of the urban localities whose residents' mortality we observe. The forest pests represented by the instrument are not known to directly harm

human health. Exclusion could nevertheless fail if outbreaks themselves affect mortality through another channel. By weakening or killing trees in the surrounding forest, outbreaks could alter fuel conditions and fire risk, creating a possible smoke pathway. Outbreaks might also disrupt local economic activity.

Independence instead concerns common causes of outbreaks and mortality. Outbreak timing is governed primarily by insect life cycles and forest conditions rather than urban socioeconomic decisions, but environmental conditions could still jointly affect outbreaks and mortality. Drought and heat stress can favor forest-pest outbreaks and also raise mortality. The same conditions may increase fire and smoke exposure or favor mosquito-borne disease.

Our main falsification addresses both classes of threat. It asks whether nearby outbreaks predict mortality in localities that begin with so little urban canopy that few potential host trees are available for pests to damage. A mortality response without a canopy response would reveal either a pathway from outbreaks to mortality that bypasses canopy loss or a common cause of outbreaks and mortality. Section 4.1 establishes the first-stage pattern needed for this test, and Section 4.5 reports the low-cover falsification.

Seasonal patterns and additional controls help distinguish among these alternatives. A mortality response concentrated in hot periods with strong solar exposure would be difficult to reconcile with slow-moving economic disruption and would also weigh against mosquito-borne disease, whose incidence in Mexico is concentrated in the rainy season rather than at the dry-season solar peak. These comparisons are less informative about climate confounding or fire and smoke because those threats can share the same dry-season timing as the canopy response. We therefore also estimate specifications that absorb climate-zone-specific annual shocks and condition on contemporaneous and antecedent weather, satellite measures of smoke, and nearby burning. Falsification tests based on nearby mistletoe and fire damage ask whether similar mortality patterns arise from generic biological stress or a non-insect ecological disturbance. Section 4.2 reports the seasonal comparisons, while Section 4.5 reports the controlled and falsification specifications.

Finally, INFyS observes forest conditions only during scheduled inventory visits, so the recorded detection date need not coincide with biological onset and may lag the beginning of an outbreak. Canopy loss associated with the outbreak may therefore begin before the recorded detection date, preventing the visit from providing a credible zero point for comparing mortality before and after exposure. Appendix B.2 traces canopy loss around the recording visit and examines both earlier exposure alignment and the exclusion of contemporaneous detections.

3.2 Testable Predictions

Our central hypothesis is that urban canopy protects health by reducing thermal exposure. The treatment margin generated by our design helps discipline the mechanism analysis: nearby outbreaks induce an incremental change in measured urban canopy relative to the locality footprint. This scale makes thermal protection operating around the affected canopy, particularly shade, the natural starting point. We use shade to encompass direct-beam shielding and the radiation-driven cooling of nearby surfaces and air. The canopy keeps shortwave radiation off people, while shaded ground and walls absorb and re-emit less heat. Trees can also cool nearby air through transpiration (Tan et al., 2018; Rahman et al., 2021). When these local effects are sufficiently extensive and spatially configured, they can accumulate into locality-wide ambient cooling (Alonzo et al., 2025). These considerations generate four predictions for the mortality response, concerning where it appears, when it arises, which deaths carry it, and who bears it, together with a complementary diagnostic based on spatial reach.

A first prediction concerns baseline canopy cover. The low-cover falsification above implies that mortality should not respond when outbreaks do not reduce urban canopy. Where outbreaks do reduce canopy, residents may adapt to a lost patch by shifting toward remaining shade, an adjustment that should be easier where canopy is abundant. The shade mechanism therefore predicts a larger mortality effect where substitute shade is scarce, a directional ordering rather than a sharp threshold.

A second prediction concerns when the mortality response should arise. Shade should protect health most when heat coincides with substantial incident solar radiation. April through June is the dry-season peak of solar exposure, whereas July through September remains hot but is substantially cloudier and receives less solar radiation. We therefore predict a larger mortality response in Q2 than in other quarters, with Q2 versus Q3 providing the sharpest seasonal comparison because both quarters are hot but differ in solar exposure. Because transpirative cooling depends on plant water availability, the wetter Q3 conditions should, if anything, be more favorable to it. Within Q2, shade predicts a weaker response in unusually overcast years, when clouds suppress the direct solar beam.

Because heat and solar radiation co-move across seasons, these comparisons test the joint heat-and-solar condition under which shade should provide protection; they do not isolate solar radiation alone. Spatial reach provides a complementary diagnostic. A locality-wide ambient-cooling pathway requires canopy loss to generate thermal changes large enough to register at the locality scale, whereas shade can change exposure around affected canopy without changing locality-average temperature.

A third prediction concerns which deaths should carry the response. A thermal protection

mechanism predicts a cardiovascular mortality response because heat stress directly strains the cardiovascular system. Respiratory mortality provides a diagnostic for pollution-related pathways. External-cause mortality provides a diagnostic for broader behavioral or economic pathways.

Finally, a fourth prediction concerns who bears the health consequences. Thermal protection should matter most for people who are physiologically more susceptible because of their age or have fewer opportunities to avoid exposure because of their work characteristics.

No single comparison identifies the mechanism. The interpretation therefore rests on the joint pattern: a mortality response where outbreaks induce canopy loss, when heat and solar exposure coincide, among causes most directly linked to thermal stress, and among populations most exposed or vulnerable to it, considered alongside evidence on the spatial reach of the thermal response.

3.3 Estimation

Our outcome is a count of deaths in locality i in year t , or in a particular quarter of that year, and the endogenous regressor is annual canopy loss measured in percentage points of locality area. We estimate mortality equations by Poisson pseudo-maximum-likelihood (PPML) (Santos Silva and Tenreyro, 2006). Because finite PPML estimates may not exist under separation, we implement these equations using the high-dimensional fixed-effects estimator of Correia et al. (2020), which detects and removes separated observations. We estimate the first stage by OLS and combine it with the mortality equation using the control-function approach of Wooldridge (2015).

For each outcome, we begin with the naïve PPML association between deaths and measured annual canopy loss, which provides the comparison reported below. The observations retained by this regression define the common outcome-specific sample for the OLS first stage, PPML reduced form, and PPML control-function equation. Sample sizes may differ slightly across outcomes, but all equations for a given outcome use the same locality-years, and each PPML equation uses the corresponding population exposure.

On this sample, we estimate the linear first stage:

$$f_{it} = \pi z_{it} + \alpha_i + \gamma_t + \eta_{it}, \tag{1}$$

where f_{it} is canopy loss in percentage points of locality area, z_{it} is the insect-outbreak instrument, and α_i and γ_t are locality and year fixed effects. We save the residuals $\hat{\eta}_{it}$.

Next, we estimate the reduced form by PPML:

$$E[d_{it} \mid \cdot] = \text{pop}_{it} \exp(\rho z_{it} + \alpha_i + \gamma_t), \quad (2)$$

where d_{it} is the death count and pop_{it} is the corresponding population at risk. Because the reduced form contains neither the estimated first-stage coefficient nor a generated residual, its inference does not inherit weak-first-stage instability. Under the independence condition in Section 3.1, the reduced form identifies the mortality effect of outbreak exposure. Attributing that effect to canopy loss additionally requires instrument relevance and the exclusion restriction. We therefore use the reduced form as the weak-instrument-robust inferential anchor.

Finally, we estimate the structural control-function equation:

$$E[d_{it} \mid \cdot] = \text{pop}_{it} \exp(\beta f_{it} + \lambda \hat{\eta}_{it} + \alpha_i + \gamma_t). \quad (3)$$

Within the maintained control-function specification, the coefficient β is the semi-elasticity with respect to measured annual Hansen loss. Including the first-stage residual controls for dependence between measured loss and unobserved determinants of mortality. This interpretation requires a linear, additively separable first stage, the residual to be sufficient for that dependence, and a log-linear conditional mean that imposes a common β within each estimation sample. The coefficient λ on the first-stage residual tests whether canopy loss not predicted by outbreaks is systematically related to mortality. If it is, the naïve Poisson association reflects those other differences as well as any effect of canopy loss.

3.4 Effect Sizes and Inference

We report the control-function estimate at the average change in measured annual canopy loss induced by the instrument. Let $\hat{\pi}$ denote this first-stage shift and define $\hat{\theta}_{CF} = \hat{\beta}\hat{\pi}$, which we express as $100[\exp(\hat{\theta}_{CF}) - 1]$. In the preferred specification, $\hat{\pi}$ is approximately 0.02 percentage points. A one-percentage-point change would therefore be roughly fifty times the identifying variation. We consequently report the model-implied mortality effect at the first-stage shift rather than extrapolate $\hat{\beta}$ to a much larger change.

Annual Hansen loss is a locality-year flow: it records newly detected loss in each year. It does not measure the full outbreak-induced reduction in standing canopy during the mortality period. Recovering that quantity would require assumptions about biological onset, whether loss before the recording visit is attributable to the outbreak, and how long lost canopy remains absent or is replaced. We therefore use measured annual loss as the en-

ogenous regressor. Accordingly, $\hat{\pi}$ is the average change in that flow while the five-year outbreak indicator is active, not total loss over an outbreak episode. Its translation into hectares describes the physical scale but does not identify a portable mortality effect per hectare or percentage point of remaining canopy.

We report the reduced form alongside the control function because the two serve different purposes. The control function gives the model-based effect at the measured canopy-loss margin and is the estimate used in the valuation. The reduced form directly estimates the outbreak-induced mortality contrast without expressing it per unit of canopy loss. It anchors inference and the main robustness comparisons. Because the model is nonlinear, evaluating the control-function estimate at the first-stage shift does not mechanically reproduce the reduced form. Their close agreement in the Results shows that both approaches imply nearly the same mortality increase at the outbreak-induced margin.

The control-function equation contains an estimated first-stage residual, $\hat{\eta}_{it}$, so standard errors from the mortality equation alone do not account for uncertainty in the first stage. We therefore use a block bootstrap at the locality level with 1,999 replications, resampling all retained observations for a locality together and re-estimating both stages in every draw. This preserves dependence within a locality over time.

In each draw, we recompute $\hat{\theta}_{CF}$ from that draw’s estimates and express it as the finite percentage effect defined above. Confidence intervals use the 2.5th and 97.5th percentiles of these effects, and the p -value is twice the smaller of the share at or below zero and the share above zero. This carries uncertainty from both stages into inference and avoids imposing symmetry on the bootstrap distribution.

Our analysis tests multiple mortality outcomes. We assign every quarterly outcome in the mechanism analysis to one family based on period and substantive role and report sharpened q -values that control the false discovery rate within each family (Anderson, 2008). Appendix B.4 lists the family assignments, explains how we treat contrasts, robustness specifications, and secondary analyses, and examines the sensitivity of reduced-form inference to alternative procedures and geographic concentration.

4 Results

4.1 Where Canopy Loss Raises Mortality

Panel A of Table 1 reports the full-sample first stage. A nearby insect outbreak increases measured annual canopy loss by 0.0247 percentage points of locality area. The effective first-stage F -statistic is 21.9 (Montiel Olea and Pflueger, 2013).

Table 1: First Stage and Annual Mortality by Baseline Canopy Cover

Baseline-cover sample	Full	Low	Medium + High	Medium	High
<i>Panel A: Measured annual canopy loss, first stage</i>					
Outbreak coefficient (π)	0.0247 (0.0053)	0.0003 (0.0012)	0.0355 (0.0070)	0.0201 (0.0063)	0.0531 (0.0118)
F-statistic	21.9	0.1	25.4	10.1	20.4
<i>Panel B: Annual all-cause mortality, Poisson control-function IV</i>					
Naive Poisson (%)	0.01	0.00	0.01	0.11	-0.11
Effect at first-stage shift (%)	0.89	—	1.63	2.72	-0.12
p-value	0.242	—	0.062	0.011	0.992
Residual coef. (λ)	-0.3563	—	-0.4532	-1.2735	0.0010
p(endogeneity)	0.255	—	0.055	0.018	0.997
<i>Panel C: Annual all-cause mortality, reduced form</i>					
Insect RF (%)	0.88	0.31	1.63	2.58	-0.12
p-value	0.255	0.812	0.054	0.016	0.926
Mean baseline canopy cover (%)	19.5	1.7	28.3	11.5	45.0
Observations	90,416	29,874	60,542	30,226	30,316
Localities	4,110	1,358	2,752	1,374	1,378

Notes: Low, Medium, and High divide localities with positive year-2000 canopy cover into terciles, with cutoffs at 4.4% and 21.8% of locality area. Full pools all three groups; Medium + High pools the upper two. Mean baseline cover weights localities equally within each column’s annual all-cause sample. Observations are locality-years, 2002–2023. Within each column, all panels use the same PPML-retained sample and locality and year fixed effects; mortality models use annual population exposure. Panel A regresses measured annual canopy loss (percentage points of locality area) on the insect-outbreak instrument. Parentheses contain analytical locality-clustered standard errors; F is the corresponding effective first-stage statistic (Montiel Olea and Pflueger, 2013). Panel B reports the Poisson control-function mortality effect at the first-stage shift, $100[\exp(\hat{\beta}\hat{\pi}) - 1]$, where $\hat{\beta}$ is the canopy-loss coefficient and $\hat{\pi}$ is the first-stage coefficient. Naive Poisson uses the same rescaling without the control function. λ is the first-stage-residual coefficient, and p(endogeneity) tests $\lambda = 0$. Control-function estimates and diagnostics are omitted for Low, where $F = 0.1$. Panel C reports PPML percentage effects of the instrument on annual all-cause mortality. Control-function effect p-values use a percentile bootstrap with 1,999 locality block-resamples that re-estimate both stages. Endogeneity-test and reduced-form p-values are analytical, with standard errors clustered by locality.

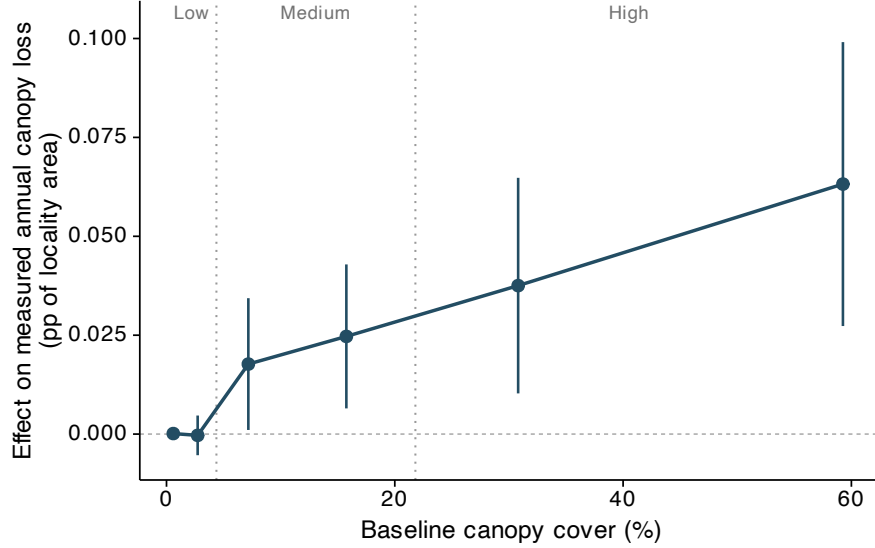


Figure 2: Effect of Nearby Insect Outbreaks on Annual Canopy Loss by Baseline Cover

Notes: The figure plots the first-stage coefficient π for measured annual canopy loss (percentage points of locality area) against baseline canopy cover. Coefficients are estimated within each of six approximately equal-sized bins of localities from separate regressions on the insect-outbreak instrument, with locality and year fixed effects and standard errors clustered at the locality level. Each point is plotted at its bin's mean baseline canopy cover. The six bins partition the annual all-cause PPML-retained sample reported in Table 1 ($N = 90,416$ locality-years across 4,110 localities). Vertical bars are 95% intervals from analytical cluster-robust standard errors. The dashed horizontal line marks zero, and the vertical lines mark the tercile cutoffs. Terciles are thirds of the baseline-cover distribution among localities with positive baseline cover; no mortality outcome enters the construction of the bins or the tercile cutoffs. The corresponding tercile-level first stage is reported in Table 1, Panel A. The national map of localities by canopy-cover tercile appears in Appendix Figure A1.

Figure 2 disaggregates the first stage across six bins of baseline canopy cover. The coefficient is small and statistically indistinguishable from zero in the two lowest bins, becomes positive in the next bin, and increases with baseline cover thereafter. The figure thus distinguishes a low-cover range with essentially no first stage from an upper range in which outbreaks increasingly predict canopy loss.

The first-stage pattern gives each cover range a distinct role. Insect outbreaks occur near low-cover localities but, consistent with the limited urban host material anticipated in Section 3.1, produce no detectable change in measured canopy loss when baseline cover is below 4.4 percent: the tercile-level coefficient is 0.0003 ($F = 0.1$). The low tercile therefore cannot identify a canopy effect, but it provides the low-cover falsification. In the remaining two-thirds of the sample, the pooled first-stage coefficient is 0.0355 ($F = 25.4$). This is the part of the cover distribution in which the design can study the mortality consequences of canopy loss.

Within this remaining two-thirds, the mechanically defined tercile partition provides the

medium–high comparison called for by the substitute-shade prediction. The boundary does not mark another change in instrument relevance: the first stage continues to rise through it. The first-stage coefficient is 0.0201 in the medium tercile ($F = 10.1$) and 0.0531 in the high tercile ($F = 20.4$). For scale, in the medium-cover sample mean annual measured loss is 0.03 percentage points of locality area and mean locality area is 5.4 km². The medium-tercile shift is therefore an increase of about 70 percent of mean annual loss, or about 0.11 hectares per outbreak-exposed locality-year. The medium tercile is the lower-cover half of this instrument-relevant sample: medium- and high-tercile localities average 11.5 and 45.0 percent baseline cover, respectively. The substitute-shade prediction therefore anticipates a larger response in the medium tercile, where remaining canopy provides fewer substitutes for lost shade. Pooling with the high tercile broadens the sample but adds localities where abundant canopy may provide substitute shade and attenuate the response. We therefore use the medium tercile as the primary sample for testing this prediction and the pooled upper two terciles to examine whether the mortality response extends across the broader instrument-relevant sample. Panels B and C test the mortality response and its predicted ordering.

In the full sample, the control-function and reduced-form annual mortality estimates are both approximately 0.9 percent and statistically indistinguishable from zero. In the low-cover falsification, nearby outbreaks are observed but measured canopy loss does not respond, and the annual mortality reduced form is 0.3 percent ($p = 0.812$), also statistically indistinguishable from zero. In the medium tercile, the control-function estimate is 2.7 percent ($p = 0.011$), equivalent to approximately 0.16 additional deaths per 1,000 person-years, and the reduced form is 2.6 percent ($p = 0.016$). Both high-tercile estimates are approximately -0.1 percent and imprecise. This ordering is consistent with the substitute-shade prediction, although the medium- and high-tercile estimates are not statistically distinguishable. Pooling the two terciles yields estimates of 1.6 percent under the control function and 1.6 percent under the reduced form ($p = 0.062$ and $p = 0.054$, respectively). The attenuation is in the direction predicted when high-cover localities are included.

Evaluated at each sample’s own first-stage shift, every naïve Poisson association is approximately 0.1 percent or smaller in absolute value. The positive medium-tercile response therefore does not reflect an unusually large observational canopy–mortality association in that sample. In the medium tercile, the first-stage residual is negative ($p = 0.018$), indicating that canopy loss not predicted by outbreaks is negatively associated with mortality. The design does not distinguish whether this pattern reflects development-related loss occurring alongside protective infrastructure, coarse measurement of partial thinning and small trees, or both.

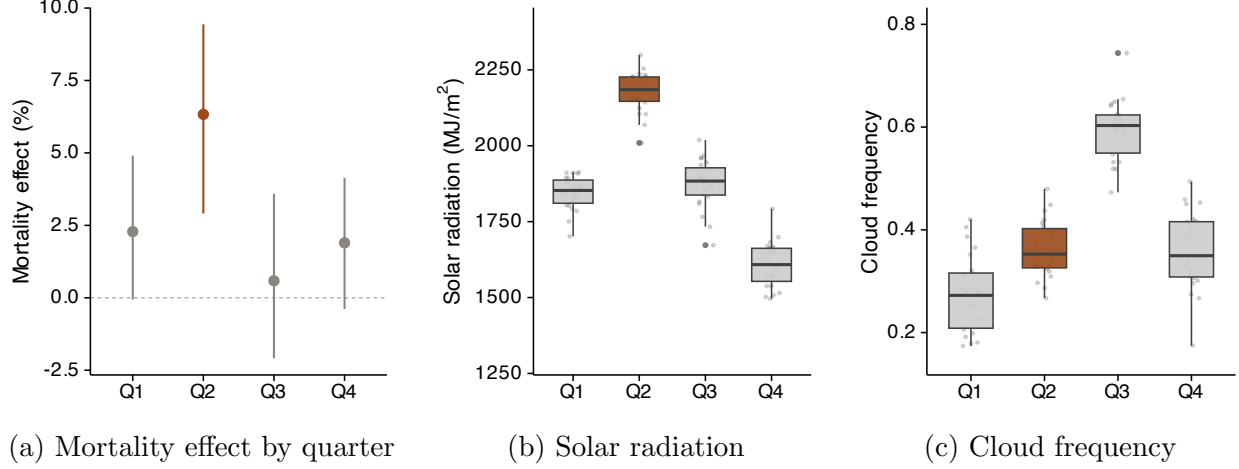


Figure 3: Mortality Effects and Solar Radiation across Seasons

Notes: All panels concern the medium canopy tercile (year-2000 cover 4.4–21.8% of locality area). Panel (a) reports quarterly all-cause mortality effects from two-step Poisson control-function IV models, expressed as percentage changes at each quarter’s first-stage shift in measured annual canopy loss. Canopy loss is measured in percentage points of locality area and instrumented with the insect-outbreak instrument. Each regression uses 30,182 locality-years from 2002–2023, locality and year fixed effects, and quarterly population exposure. Vertical bars are 95% percentile-bootstrap confidence intervals from 1,999 locality block-resamples that re-estimate both stages. Paired contrasts use the same draws, with percentile-bootstrap p-values: Q2 minus Q1 is 4.0 percentage points ($p = 0.014$), Q2 minus Q3 is 5.7 ($p < 0.001$), and Q2 minus Q4 is 4.4 ($p = 0.002$). Panel (b) shows ERA5-Land surface solar radiation as quarterly totals in MJ/m². Panel (c) shows MODIS cloud frequency, the quarterly share of daily 1-km observations at the locality centroid classified as cloudy or mixed. For each quarter, these box plots summarize 23 year-specific means across medium-cover localities, 2001–2023; points show the individual year means. Q1–Q4 denote calendar quarters; Q2 is shown in rust and the others in grey. The dashed line in Panel (a) marks zero. Appendix Table A3 reports the mortality estimates and Table A4 reports climate summaries and variable definitions.

On observed settlement characteristics, the medium-cover and full-sample estimation samples are similar. Mean population, locality area, population density, built-surface share, outbreak prevalence, and mortality counts are comparable across them (Appendix Table A2). Baseline canopy differs by construction, and mean annual canopy loss is lower as a share of locality area in the medium-cover sample.

4.2 When Canopy Loss Raises Mortality

We begin with the medium-cover tercile, the primary sample identified above. The 2.7% annual estimate masks considerable seasonal heterogeneity. Figure 3a shows that the Q2 control-function effect is 6.3% ($p = 0.001$), equivalent to approximately 0.36 additional deaths per 1,000 person-years in that quarter. The corresponding reduced form is 6.2% ($p < 0.001$). The Q1, Q3, and Q4 point estimates are smaller and individually imprecise. Paired comparisons confirm that Q2 exceeds every other quarter ($p \leq 0.014$).

Q3 provides the most informative seasonal comparison because Q2 and Q3 are both hot

but differ sharply in accompanying solar exposure and moisture. Using the daily-average temperature bins standard in the heat–mortality literature, Q2 averages approximately 11 days at or above 28°C and Q3 approximately 9, with days at or above 32°C rare in both quarters. Q3 is substantially cloudier and wetter: surface solar radiation is approximately 300 MJ/m² lower than in Q2, mean cloud frequency rises from 0.36 to 0.59, and relative humidity rises from 59% to 77% (Figures 3b and 3c; Appendix Table A4). These seasonal differences are geographically widespread: comparing locality-specific averages over 2001–2023, Q3 receives less solar radiation in 99.5% of sampled localities and has higher cloud frequency in 98.0%. Wetter Q3 conditions should, if anything, be more favorable to transpirative cooling and rainy-season mosquito transmission. Yet the control-function estimates give a Q3 effect of 0.6% ($p = 0.637$) and a Q2–Q3 difference of 5.7 percentage points ($p < 0.001$). This pattern weighs against a response to heat independent of solar exposure, transpirative cooling as the primary mechanism, and a rainy-season vector pathway. It instead matches the prediction that shade protects most when heat coincides with substantial solar exposure.

Variation within Q2 provides a second test. We classify Q2 locality-years using cloud-frequency and solar-radiation deviations from each locality’s own seasonal norm. The mortality estimate is small and imprecise in unusually overcast Q2 years but positive and statistically significant under both normal and unusually sunny conditions (Appendix Table A6). When the latter two groups are pooled, their difference from overcast years is suggestive ($p = 0.066$ when years are binned by solar radiation and $p = 0.067$ when binned by cloud frequency). The comparison does not isolate solar exposure from heat, but it holds the quarter fixed. The resulting pattern weighs against other explanations specific to Q2 and provides suggestive within-Q2 support for the shade interpretation.

The seasonal signature also extends across the broader instrument-relevant sample defined in Section 4.1. Pooling the medium- and high-cover terciles roughly doubles the Q2 estimation sample. The pooled Q2 control-function estimate is 4.5% ($p = 0.001$), and the corresponding reduced form is also 4.5% ($p < 0.001$). For the high-cover tercile alone, the Q2 reduced form is 1.8% ($p = 0.259$). The pooled Q3 reduced form remains statistically indistinguishable from zero, and the Q2–Q3 contrast remains positive and statistically significant ($p < 0.001$; Appendix Table A5, Panel B). Broadening the sample attenuates the estimate in line with the cover pattern in Section 4.1, but does not alter its seasonal signature. The positive annual estimate and the absence of negative estimates in the other quarters likewise provide no evidence of short-run mortality displacement.

The seasonal evidence points to shade, but it does not establish the spatial scale at which thermal protection operates. Shade can alter pedestrian exposure around affected canopy without measurably changing locality-average temperature. Small-scale field measurements

show that shade can reduce physiologically equivalent temperature by approximately 13°C relative to sun (Rahman et al., 2021), illustrating this distinction. Because the design links mortality to localities rather than to individuals’ use of affected spaces, it does not identify how many residents lose shade exposure or by how much their thermal exposure changes.

We can, however, ask whether outbreaks generate enough locality-wide heating to explain the 6.2% Q2 mortality reduced form. Nearby outbreaks lower the Q2 mean of daily maximum temperature by 0.05°C rather than raise it. Even when we treat the most adverse displacement permitted by the 95% confidence interval as warming and pair it with the upper end of the temperature–mortality response, locality-wide air-temperature changes can explain at most 6.5% of the mortality response; bounds based on counts of days with daily average temperature at or above 28°C or 32°C are less than 2%. Because the calculation compares reduced forms of the same instrument, it does not depend on the Hansen measure capturing the full canopy response. Appendix C provides the complete calculation across thermal measures and canopy-cover samples. These bounds rule out locality-wide temperature as the principal channel and, together with the seasonal evidence, point to shade operating around affected canopy.

4.3 Who Bears the Cost: Causes and Distributional Incidence

We next examine how the mortality response varies across causes of death and population groups. Figure 4 reports Q2 and non-Q2 estimates for cardiovascular, respiratory, and external causes; for children, working-age adults, and older adults; and, within working age, by insurance status. The cause groups probe mechanism, while age and insurance status capture differences in physiological vulnerability and opportunities to avoid exposure.

Cardiovascular mortality rises by 7.3% in Q2 ($p = 0.001$), and the Q2 estimate exceeds its non-Q2 counterpart by 6.9 percentage points ($p < 0.001$). We detect no clear response in respiratory or external-cause mortality, although both estimates are imprecise. The smoke controls and falsification evidence in Section 4.5 more directly weigh against smoke-related air pollution and broad economic disruption as primary explanations.

Mortality rises in Q2 by 6.1% among adults aged 65 and older ($p = 0.002$) and by 4.5% among working-age adults ($p = 0.024$), with no detectable response among children. The response is therefore not confined to physiological vulnerability among older adults. The working-age response accords with Mexico-specific evidence that heat mortality extends to younger adults and that occupational exposure may matter (Wilson et al., 2024; Bressler et al., 2026), although the age estimates do not identify where exposure occurs.

Within working age, we ask whether mortality is greater among people with greater

exposure or fewer means of avoiding or managing heat. Insurance status is an informative but imperfect proxy for both exposure and capacity to adapt. Formal coverage in our data consists mainly of employment-linked social-insurance institutions, whereas the informal group comprises uninsured people and enrollees in Seguro Popular, a public program not tied to formal employment. Insurance status therefore captures a bundle of labor-market and household conditions, including outdoor or poorly cooled work, resources for private cooling, baseline health, and access to care.

Q2 mortality rises by 8.8% among informally insured working-age adults ($p = 0.035$), compared with 2.6% among the formally insured ($p = 0.436$). The informal-minus-formal Q2 gap is 6.1 percentage points and imprecisely estimated ($p = 0.063$). Because this distributional comparison concerns incidence across groups rather than seasonal timing, we also use mortality from all four quarters. The annual gap is similar in magnitude at 5.1 percentage points and statistically distinguishable from zero ($p = 0.016$). Annual occupation estimates point in the same direction: the estimate is larger for outdoor than indoor occupations, although the difference is not statistically distinguishable and the occupation data are too incomplete and coarsely classified for a definitive test (Appendix Table A8). Together, these results suggest regressive incidence, but the design cannot determine how much of the insurance gap reflects workplace exposure, cooling resources, baseline health, or access to care.

The main subgroup results extend beyond the medium tercile. When medium- and high-cover localities are pooled, the cardiovascular, older-adult, and working-age reduced forms remain positive and statistically significant, while the informal-insurance estimate remains positive but less precise (Appendix Table A5).

Taken together with the cover, seasonal, and spatial evidence above, the cause-of-death and distributional patterns further support lost shade as the main mortality channel at the identified margin.

A separate place-based equity question concerns where existing canopy is located. The year-2000 Social Lag Index measures locality deprivation. Mean baseline canopy rises across its terciles, from 10.3% among the least-deprived localities to 27.0% among the most-deprived. The relationship remains positive when comparing localities within the same municipality and accounting for differences in local built development (Appendix Table A9). This is the opposite of the familiar pro-rich canopy gradient documented elsewhere (McDonald et al., 2021). The canopy gradient and the mortality-incidence estimates describe distinct margins: the former concerns where canopy is located, while the latter concerns which residents appear more vulnerable to its loss. The design does not identify their intersection. The descriptive gradient nevertheless highlights why canopy conservation matters

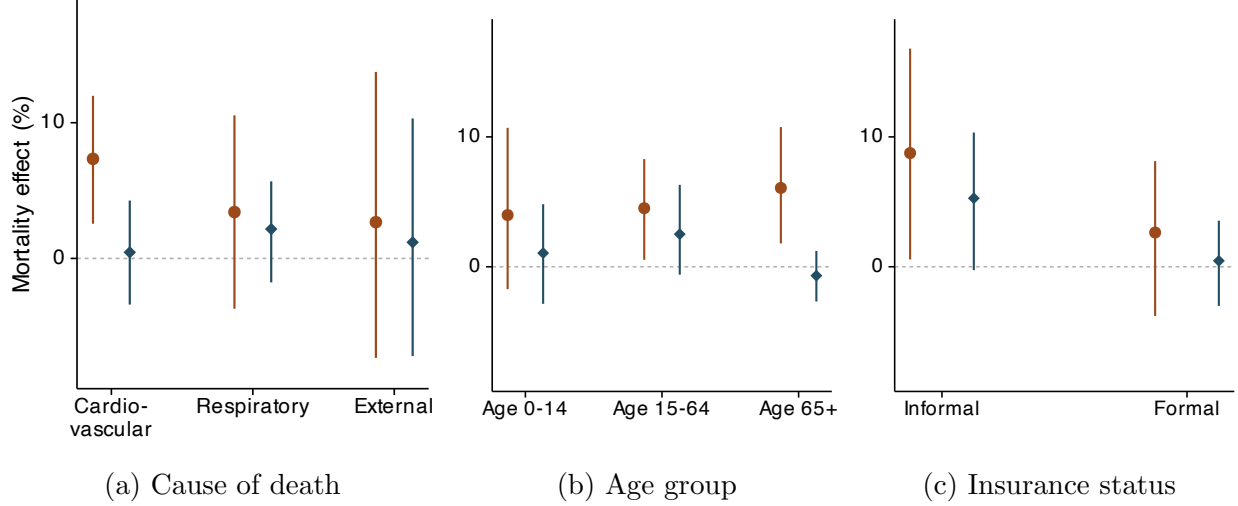


Figure 4: Mortality Effects by Cause and Population: Q2 versus Non-Q2

Notes: Each panel reports two-step Poisson control-function IV estimates for mortality in Q2 (April–June) and non-Q2 (Q1, Q3, and Q4 pooled), expressed as percentage changes at the corresponding outcome-specific first-stage shift in measured annual canopy loss. Canopy loss is measured in percentage points of locality area and instrumented with the insect-outbreak instrument. Regressions use locality-years from 2002–2023 in the medium canopy tercile (year-2000 cover 4.4–21.8% of locality area), locality and year fixed effects, and population exposure matched to each outcome’s period and age group. Panel (c) covers ages 15–64. Informal includes uninsured people and enrollees in non-employment-linked public programs, including Seguro Popular; Formal includes employment-linked and private coverage. Vertical bars are 95% percentile-bootstrap confidence intervals from 1,999 locality block-resamples that re-estimate both stages. The paired Informal-minus-Formal Q2 gap is 6.1 percentage points, with a percentile-bootstrap p-value of 0.063. Rust circles denote Q2; navy diamonds denote non-Q2. Horizontal reference lines mark zero, and all panels share a common vertical scale. Appendix Table A7 reports exact estimates, p-values, and outcome-specific observation counts.

for equity in Mexico: more-deprived localities already hold substantial existing canopy, so policy should consider its conservation as well as where new canopy is provided.

4.4 The Cost of Attributable Deaths

We translate the preferred Q2 all-cause estimate into attributable mortality in the medium-cover sample. Because the cause and demographic groups in Section 4.3 overlap, we use the all-cause estimate rather than summing subgroup effects. The 6.33% Poisson control-function estimate implies approximately 2,647 attributable April–June deaths across the 2,532 outbreak-exposed locality-years observed from 2002 through 2023. This corresponds to an average of about 120 deaths per year, or roughly one additional death per outbreak-exposed locality-year.

To assess the economic magnitude of these deaths, Table 2 values the annualized average of 120 deaths under three alternative valuation assumptions. The main transfer reconstructs Banzhaf (2022)’s U.S. 2019 VSL distribution and transfers it to Mexico using PPP GDP per capita and an income elasticity of one, consistent with Viscusi and Masterman (2017) and the default in OECD (2025).

Table 2: Annualized Mortality Cost under Alternative Valuation Assumptions

	U.S. transfer, $\eta = 1$	U.S. transfer, $\eta = 1.5$	LMIC transfer, $\eta = 1$
Mean VSL (Mexico)	\$2.82	\$1.57	\$1.87
Annualized mortality cost	\$339.1	\$188.6	\$225.5
90% uncertainty interval	[\$87.9, \$661.5]	[\$48.9, \$368.0]	–

Notes: Values are millions of 2022 U.S. dollars; VSL denotes the value of a statistical life. Columns 1 and 2 transfer Banzhaf (2022)’s U.S. 2019 VSL distribution to Mexico using PPP GDP per capita and income elasticities $\eta = 1$ and $\eta = 1.5$, respectively, with CPI-U adjustment to 2022. Column 3 scales the OECD (2025) low- and middle-income-country (LMIC) mean VSL to Mexico using PPP GDP per capita and $\eta = 1$. All columns multiply mean VSL by the same annualized attributable mortality: approximately 120 Q2 (April–June) deaths per year in the medium-cover sample (year-2000 cover 4.4–21.8% of locality area). Annualized attributable mortality equals 44,453 observed Q2 deaths across 2,532 outbreak-exposed locality-years times $e/(e + 100)$, divided by 22 years (2002–2023), where $e = 6.33$ is the Q2 all-cause control-function percentage effect. The 90% uncertainty intervals in Columns 1 and 2 come from independently pairing 1,000,000 VSL draws with mortality-effect draws from 1,999 locality block-resamples that re-estimate both stages. VSL draws are not truncated at zero; 0.61% are negative. Column 3 uses a fixed VSL, so an interval would reflect mortality uncertainty alone and is omitted.

Under this transfer, the mean VSL is \$2.82 million in 2022 dollars, giving an annualized mortality cost of \$339 million within the study sample. Combining independent VSL and mortality-effect draws gives a 90% uncertainty interval of \$88–662 million. Most of this dispersion comes from the VSL literature: VSL draws account for 72.2% of the simulated variance, mortality draws for 23.2%, and their interaction for 4.6%. More conservative

transfers preserve the order of magnitude: using an income elasticity of 1.5 gives \$189 million, while scaling the OECD’s 2025 low- and middle-income-country mean VSL to Mexico’s PPP GDP per capita gives \$226 million.

We do not express this mortality cost per hectare, for two reasons. First, translating the first-stage shift into hectares describes an annual flow of newly detected Hansen loss. Recovering the outbreak-induced reduction in standing canopy during April–June from that flow would require the additional assumptions described in Section 3.4. Appendix B.2.2 documents cumulative post-recording Hansen loss, but it does not resolve those assumptions. Hansen also does not directly measure partial thinning or losses of individual small trees. Dividing the cost by measured annual hectares would therefore treat one year’s recorded flow as the reduction in standing canopy relevant to April–June mortality. The resulting ratio would be a design-specific normalization by the Hansen measure, not an identified per-hectare value or a bound on one. Second, even if that reduction were observed, a portable per-hectare value would require each lost hectare to remove equivalent protection regardless of its location, spatial configuration, and available substitute shade. It would also require the mortality response to scale proportionally beyond the finite instrument-induced change. These limits restrict per-hectare extrapolation across projects and to new planting, not the relevance of the estimate for incremental decisions about whether existing canopy remains in place.

This mortality accounting is also narrower than a complete benefit-cost calculation. The \$339 million already incorporates the avoidance and private adaptation residents undertake. Previous research shows that heat exposure also affects activity, labor supply, productivity, and learning (Graff Zivin and Neidell, 2014; Park et al., 2020; Somanathan et al., 2021). To the extent that retaining existing shade protects these outcomes, mortality is only one component of its potential gross benefit. We do not estimate those margins here or offset them against costs and canopy disservices, so this amount is not a bound on net welfare. Even with these limits, the accounting shows that mortality losses at the identified conservation margin are economically consequential.

4.5 Robustness

The medium-cover Q2 all-cause reduced form is 6.15% ($p < 0.001$). The reduced form anchors the robustness analysis because it directly estimates the outbreak-induced mortality contrast without dividing by the first stage or relying on a generated control-function residual. We organize the discussion around exclusion, independence, timing, instrument construction, and inference. Figure 5 summarizes selected falsification, control, and timing

estimates; Appendix B reports the complete results and diagnostics.

We begin with the low-cover falsification. The outbreak instrument is active in 6.4% of low-cover locality-years, compared with 8.4% in medium cover, but generates essentially no measurable change in low-cover urban canopy. This comparison therefore retains outbreak exposure but has no measurable canopy first stage. The Q2 mortality reduced form is 1.34% ($p = 0.418$; 95% CI $[-1.87, 4.64]$). Although imprecise around zero, the interval excludes a low-cover response as large as the 6.15% medium-cover benchmark.

This comparison bears on both identifying assumptions. Environmental conditions that jointly predict outbreaks and mortality could still generate a low-cover mortality response, as could direct responses to nearby detections that do not require urban canopy loss. The low-cover result therefore weighs against both types of threat, although it does not rule out smaller effects.

Mistletoe and fire provide complementary falsifications for generic biological stress and non-insect ecological disturbance. Their mortality reduced forms are 0.91% ($p = 0.660$) and -1.63% ($p = 0.378$), respectively, and their canopy first-stage F -statistics are 6.1 and 0.4. Their limited first-stage relevance is why we interpret them as tests of direct disturbance channels rather than as attenuated canopy-effect estimates.

We next examine climate and smoke directly. Drought or other environmental conditions could contribute both to forest outbreaks and mortality, threatening independence. By weakening or killing nearby forest trees, outbreaks could also alter fuel conditions and fire risk; the resulting smoke could affect mortality directly, bypassing urban canopy loss and threatening exclusion.

To assess the empirical relevance of these threats, Appendix Table B2 reports the instrument's associations with contemporaneous and lagged climate, drought, fire, and aerosol measures. The contemporaneous pattern does not resemble an acute hot-and-dry mortality shock: none of the standardized movements exceeds 0.1 standard deviations, and outbreak-exposed locality-years are cooler, less sunny, more humid, and wetter. Associations with lagged drought and aerosol measures nevertheless motivate specifications that condition on both current and antecedent conditions.

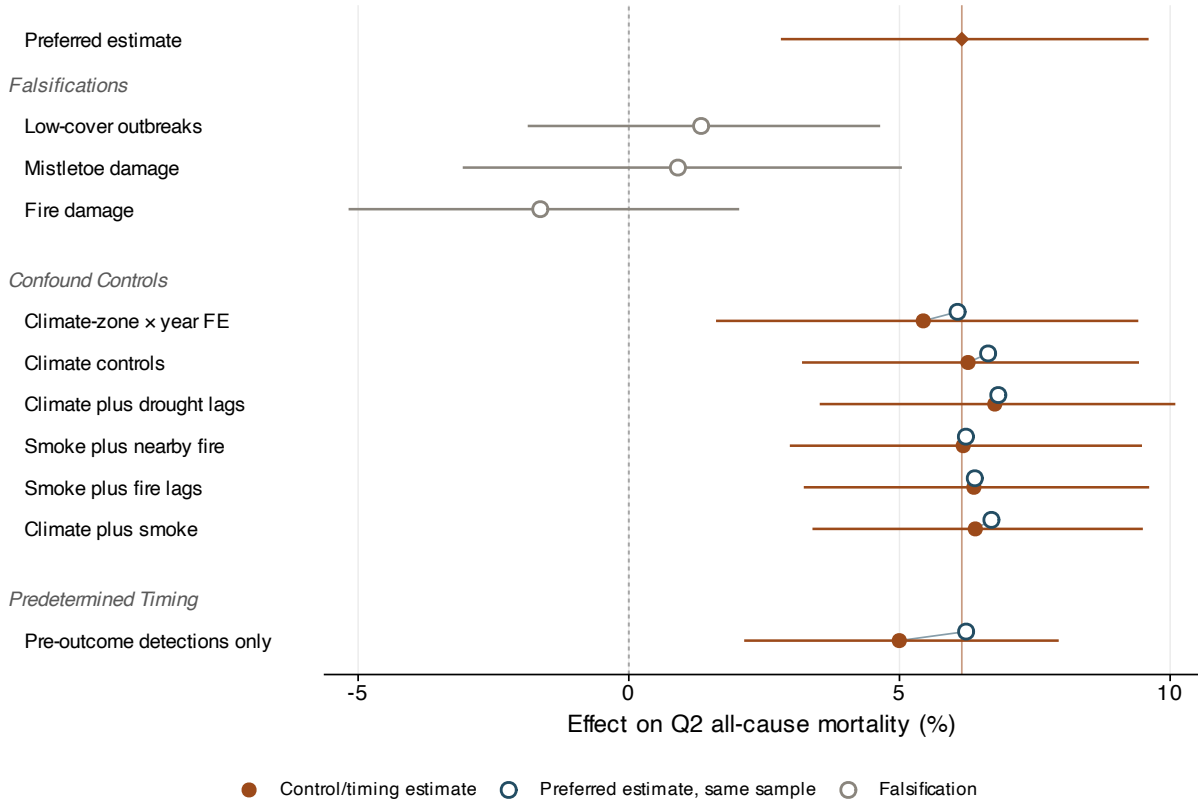


Figure 5: Falsification and Robustness Tests for the Q2 All-Cause Reduced Form

Notes: Points report PPML reduced-form percentage effects on Q2 (April–June) all-cause mortality using locality-year observations from 2002–2023. Unless indicated otherwise, regressions use the insect-outbreak instrument, the medium-cover tercile (year-2000 cover 4.4–21.8% of locality area), locality and year fixed effects, and quarterly population exposure. The diamond and solid vertical line mark the preferred estimate. The low-cover falsification uses localities with baseline cover below 4.4%; the mistletoe and fire falsifications replace the insect-outbreak instrument with the corresponding disturbance indicator. In the control block, the climate-zone specification replaces year effects with climate-zone-by-year effects. The predetermined instrument uses qualifying detections in the five years preceding the mortality year. For each control or timing row, the open navy circle shows the preferred estimate on exactly the same observations; thin segments connect the pair. Horizontal bars are analytical 95% confidence intervals, with standard errors clustered by locality; the open navy benchmarks are shown without intervals. The dashed vertical line marks zero. Appendix Table B1 reports specification details, exact estimates, p-values, sample sizes, and falsification first-stage diagnostics.

Introducing these controls changes the estimation sample slightly. We therefore pair each controlled estimate with the uncontrolled reduced form estimated on exactly the same observations, shown by the hollow navy circles in Figure 5. The controlled estimates remain between 5.44% and 6.76%, with $p \leq 0.005$ in every specification, compared with exact-sample benchmarks between 6.07% and 6.83%. Absorbing climate-zone-by-year shocks produces the largest movement, from 6.07% to 5.44%; every other controlled estimate is within 0.38 percentage points of its benchmark. Conditioning on measured climate, drought, fire, and smoke therefore leaves the Q2 response largely unchanged (Appendix Table B1).

We next address two timing limitations. First, Hansen canopy loss and outbreak exposure are measured annually, while the mortality outcome covers April through June. These annual measures therefore cannot establish the ordering of exposure, measured loss, and mortality within the same calendar year. Restricting the instrument to outbreaks recorded before the mortality year removes same-year detections and lowers the exact-sample estimate from 6.23% ($p < 0.001$) to 5.00% ($p < 0.001$).

Second, scheduled INFyS visits may record outbreaks after biological onset. We open the exposure window one to three years before the recording visit (Appendix B.2.3). These alignments produce Q2 estimates between 6.24% and 6.59% (all $p < 0.001$), against 6.15% when the instrument begins in the recording year. The two exercises change different timing margins: predetermined coding removes contemporaneous detections, whereas earlier alignment allows biological damage to precede a later recording. Predetermined coding attenuates the estimate but leaves it positive and precisely estimated; the earlier alignments remain close to the recording-year estimate. The positive Q2 response therefore persists under both treatments of timing uncertainty.

To test whether the result depends on how the outbreak instrument is constructed, we re-estimate the reduced form across 81 combinations of the spatial radius, damage threshold, qualifying-plot share, and carry-forward window. Every Q2 point estimate is positive. The preferred estimate lies near the upper end of the 2.96–6.27% interquartile range, and imprecision is concentrated among definitions with weaker first stages (Appendix Figure B2).

The preceding exercise varies the construction of outbreak exposure. A separate first-stage check asks whether instrument relevance and the low-versus-medium cover pattern depend on the Hansen canopy measure. An alternative continuous canopy product reproduces both the absent low-cover first stage and the positive medium-cover relationship (Appendix Table B3).

We next address weak-instrument uncertainty. The medium-cover Q2 effective first-stage F -statistic is 10.18, low enough to warrant caution. In a linear 2SLS specification, the estimated effect is 7.0%, and the weak-instrument-robust Anderson–Rubin 95% confidence

set is [1.8%, 20.7%]. Its lower endpoint remains above zero, while its width makes the first-stage uncertainty visible.

Turning to spatial inference, municipality and state clustering and the state wild bootstrap produce p -values no larger than 0.011. The 28 Conley specifications combine seven spatial cutoffs (25–300 km) with four serial-correlation windows (0–7 years), producing p -values from 0.004 to 0.030; permutation inference yields $p = 0.001$. Dropping each state in turn leaves all 31 PPML estimates positive with $p < 0.05$ (Appendix Table B4 and Figure B3).

Finally, we account for multiple testing within the outcome families defined in Appendix B.4. Sharpened false-discovery-rate corrections leave the Q2 all-cause hypothesis at $q = 0.005$. Four Q2 subgroup estimates also have $q \leq 0.037$: cardiovascular mortality, adults aged 65 and older, working-age adults, and informally insured working-age adults. The remaining Q2 outcomes and all non-Q2 subgroup outcomes have q -values above 0.05.

Together with the seasonal and external-cause evidence in Sections 4.2–4.3, these checks make the Q2 response difficult to attribute to direct outbreak pathways or measured climate and smoke, and unlikely to be an artifact of recording conventions, instrument construction, or inference choices. The designated falsifications yield substantially smaller and imprecise estimates, while no non-Q2 subgroup outcome survives multiple-testing adjustment.

5 Conclusion

Cities around the world are increasingly turning to tree planting and canopy conservation as part of their response to rising urban heat. Although extensive evidence shows that trees and forests can moderate heat, much less is known about the health value of preserving established canopy at the incremental scale relevant for urban conservation decisions. We provide causal evidence on this margin for Mexican cities. In the medium-cover sample, we find that the instrument-induced canopy loss raises April–June mortality by 6.3 percent. The converging evidence points to shade as the main thermal service at this margin. Conservation of established canopy is therefore a form of climate adaptation whose health value depends on where shade is available and when heat and solar exposure coincide.

The localized spatial scale of this benefit matters for valuation. Assessments based on ambient temperature capture it only to the extent that canopy changes temperature at the spatial scale being measured. In our setting, changes in locality-wide temperature can account for at most 6.5 percent of the mortality response. Hedonic approaches capture the benefit only to the extent that it is capitalized into sale or rental prices, while shade can protect people whose exposure is not tied to owning or occupying nearby property, including

while they work or move through shared spaces. By observing mortality among locality residents, our design requires neither signal. Methods relying exclusively on either signal may therefore understate mortality benefits that operate below the resolution of temperature measures or accrue beyond property occupants. The April–June estimate implies an annualized mortality cost of \$339 million in the medium-cover sample. It is neither the portion missed by other methods nor an estimate of net welfare. This valuation applies to the finite instrument-induced margin and is not a portable per-hectare value. Even with these limits, the estimate indicates that the mortality consequences of lost shade at the identified margin are substantial.

The mortality value of canopy conservation also varies with environmental conditions and available substitutes. The patterns across seasons, cloud conditions, and canopy-cover groups point to a substitution principle: conservation matters most when heat and solar exposure coincide and alternative shielding from direct solar radiation is limited. The seasonal contrast provides the strongest evidence for this principle. The cloud and canopy-cover patterns point in the same direction, although the medium- and high-cover estimates are not statistically distinguishable. Uniform valuations would therefore obscure this heterogeneity and risk understating protection under the conditions in which shade is most consequential.

The distributional results also raise an equity question. Existing canopy is descriptively more abundant in more-deprived localities, so part of the conservation value lies in preserving an environmental asset already present in those places rather than relying only on new planting. Separately, the largest working-age mortality point estimate falls on people outside formal employment-based insurance. The subgroup differences are imprecise, and the design does not connect individual incidence to the spatial distribution of canopy. Even so, these patterns suggest that canopy conservation has equity implications both because of where existing trees are located and because of who appears most vulnerable to their loss.

These estimates are identified from canopy loss induced by nearby outbreaks in Mexican urban localities. Because the evidence points to shade, the findings are informative for incremental conservation decisions that determine whether established canopy remains. How the magnitude varies with the cause, timing, persistence, scale, or spatial configuration of canopy loss, as well as across settings with different climates, labor-market conditions, or access to private cooling, remains an open empirical question. The evidence speaks to conserving established canopy, not to the benefits of planting new trees. Planting depends on siting, survival, maturation, and the eventual distribution of shade. Although we do not estimate the benefits of a planting program, the results indicate where and under what conditions its eventual shade could be especially valuable. Climate change may make this conservation margin increasingly important as high-temperature conditions become more

frequent, particularly where they coincide with strong solar exposure, but our estimates do not project future benefits. At the incremental margin studied here, conserving established canopy preserves shared protection against mortality risk when heat and solar exposure coincide.

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Appendix A Additional Tables and Figures

A.1 Data and Instrument Construction

Tables A1–A2 document the source data and compare the full and medium-cover annual all-cause estimation samples.

Table A1: Data Sources

Dataset	Variables used	Resolution / unit	Coverage	Source
<i>Panel A. Canopy cover and forest health</i>				
Hansen Global Forest Change	Baseline canopy cover (2000) and annual canopy loss	30 m	2000–2023	Hansen et al. (2013, 2023)
NASA CMS tree cover	Continuous tree cover and annual net decline (alternative measure)	30 m	2000–2019	Tran et al. (2025)
INEGI urban locality boundaries	Locality polygons for spatial aggregation	locality polygon	2010	INEGI (2010)
INFyS / CONAFOR forest inventory	Tree-level insect damage in forest plots (instrument)	forest plot	2004–2020	CONAFOR (2004–2020)
<i>Panel B. Mortality and population</i>				
INEGI vital statistics	Universe of death records (locality, month, age, sex, ICD-10 cause, occupation, insurance)	individual record	2002–2023	INEGI (2023)
INEGI censuses & counts	Population by locality, age, and sex	locality	2000–2020	INEGI (2021)
CONAPO population projections	Municipal annual growth rates for intercensal interpolation	municipality	2000–2023	CONAPO (2024)
<i>Panel C. Climate and environmental variables (seasonal context, confound controls, and thermal bounds)</i>				
MODIS MOD09GA	Quarterly cloud frequency	1 km	2001–2023	Vermote and Wolfe (2021)

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Table A1 (continued)

Dataset	Variables used	Resolution / unit	Coverage	Source
ERA5-Land re-analysis	Surface solar radiation, air and dewpoint temperature (relative humidity derived), precipitation	0.1° grid	2001–2023	Muñoz-Sabater (2019); Muñoz-Sabater et al. (2021)
ERA5-HEAT re-analysis	Universal Thermal Climate Index, mean radiant temperature	~28 km	2001–2023	Di Napoli et al. (2021); Copernicus Climate Change Service (C3S)
TerraClimate	Palmer Drought Severity Index	~4 km	2001–2023	Abatzoglou et al. (2018)
Köppen-Geiger climate classification	Long-run climate zone	1 km	1991–2020	Beck et al. (2023a,b)
MODIS MOD11A1	Daytime land-surface temperature	1 km	2001–2023	Wan et al. (2021)
MODIS MCD19A2 (MAIAC)	Aerosol optical depth at 550 nm	1 km	2001–2023	Lyapustin et al. (2018); Lyapustin and Wang (2022)
MODIS FIRMS	Active-fire detection-days	1 km	2001–2023	Davies et al. (2009); Giglio et al. (2016); NASA FIRMS (2021)
<i>Panel D. Settlement and deprivation (descriptive)</i>				
Global Human Settlement Layer	Built-up surface share of locality area	100 m	2000	Pesaresi and Politis (2023)
CONEVAL Social Lag Index	Multidimensional deprivation index (higher = more deprived)	locality	2000	CONEVAL (2000)

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Table A1 (continued)

Dataset	Variables used	Resolution / unit	Coverage	Source
<i>Panel E. Mortality valuation inputs</i>				
VSL meta-analysis and supplement	U.S. 2019 VSL mixture, component distributions, and weights	meta-study component	target year 2019	Banzhaf (2022)
BLS Consumer Price Index	Annual-average CPI-U for dollar conversion	national annual index	2019, 2022	U.S. Bureau of Labor Statistics (2026)
World Development Indicators	PPP GDP per capita for cross-country and intertemporal transfer	country-year	U.S. 2019; Mexico 2022	World Bank (2026)
OECD mortality-risk valuation	Income-elasticity guidance and low/middle-income-country benchmark	policy meta-analysis	2022 USD	OECD (2025)

Notes: “Resolution / unit” gives the native spatial resolution for raster products or the unit of observation for administrative and valuation data. Section 2 describes construction of the locality-level variables. Most gridded climate and environmental series are sampled at locality centroids; MODIS land-surface temperature is averaged across locality polygons, active-fire detections are summed within 10 km of each locality centroid, and the Köppen-Geiger classification is assigned at the centroid. Panel E inputs are used only for the mortality valuation in Table 2.

Table A2: Summary Statistics: Full and Medium-Cover Samples

	Full sample		Medium-cover sample	
	Mean	SD	Mean	SD
<i>Panel A: Settlement, canopy, and treatment</i>				
Population	19,960.24	92,631.65	20,201.71	91,194.63
Canopy loss (pp of locality area)	0.0512	0.3554	0.0297	0.2646
Canopy loss (m ²)	3,359.65	53,747.10	3,406.67	49,433.41
Baseline canopy cover (2000, 0–1)	0.1948	0.2134	0.1147	0.0502
Locality area (km ²)	5.15	14.07	5.40	13.03
Population density (per km ²)	2,672.81	2,566.43	2,623.95	2,397.57
Built-surface share (2000, 0–1)	0.1404	0.0726	0.1460	0.0715
Insect outbreak (0–1 indicator)	0.0872	0.2821	0.0838	0.2770
<i>Panel B: Mortality outcomes</i>				
Annual deaths	116.40	568.61	120.87	584.57
Q2 deaths (Apr–Jun)	27.10	133.99	28.39	140.14
Cardiovascular deaths	28.61	142.28	30.19	152.20
Respiratory deaths	9.90	51.53	10.16	51.85
External-cause deaths	10.97	48.97	11.01	42.81
Deaths age 0–14	6.52	28.71	6.35	26.62
Deaths age 15–64	44.73	220.91	45.35	219.34
Deaths age 65 and older	65.15	327.84	69.17	346.46
Outdoor-occupation deaths	7.98	34.32	8.18	34.39
Indoor-occupation deaths	14.01	72.25	14.25	72.09
Unclassified working-age deaths	22.74	116.55	22.93	114.96
Informal-insurance deaths (15–64)	19.23	87.56	19.91	88.59
Formal-insurance deaths (15–64)	20.41	117.06	20.40	117.94

Notes: Means and standard deviations over locality-years in the annual all-cause estimation samples, 2002–2023. The Full columns contain 90,416 observations across 4,110 localities. The Medium-cover columns contain the corresponding sample restricted to baseline canopy cover of 4.4–21.8% of locality area: 30,226 observations across 1,374 localities. Every mortality row except Q2 deaths reports an annual count; Q2 deaths cover April–June. Canopy loss is an annual flow reported in percentage points of locality area and square meters. Baseline canopy and built surface are shares of locality area reported on a 0–1 scale. The insect-outbreak instrument is a binary indicator. Occupation and insurance outcomes cover ages 15–64. Outdoor occupation combines primary-sector and sales/services workers; indoor occupation combines technical/manual and professional workers. Unclassified denotes working-age deaths not assigned to either occupation group. Informal insurance comprises the uninsured and Seguro Popular; formal insurance comprises IMSS, ISSSTE, military, PEMEX, and private coverage.

A.2 Where Effects Arise

Figure A1 shows the geographic coverage of the three baseline-canopy groups.



Figure A1: Analysis Localities by Baseline Canopy Cover Tercile

Notes: Dots mark the 4,110 localities in the full annual all-cause estimation sample, 2002–2023. Colors distinguish Low, Medium, and High baseline canopy cover, defined as the share of locality area covered by canopy in 2000. Fixed cutoffs of 4.4% and 21.8% separate the groups, which contain 1,358, 1,374, and 1,378 localities, respectively. No urban locality in Baja California enters the sample because none has Hansen-detected baseline canopy.

A.3 When Effects Arise

This subsection develops the seasonal evidence in Section 4.2. Table A3 reports the quarterly all-cause estimates behind Figure 3. Table A4 compares Q2 and Q3 thermal conditions, Table A5 extends the Q2 control function and the reduced forms to the pooled top two terciles, and Table A6 examines variation within Q2.

Table A3: Seasonal All-Cause Mortality Effects: Medium Canopy Tercile

Specification	N	F	Naive (%)	CF effect (%)	p-value	λ	p(endog)
Q1 (Cool/Dry)	30,182	9.8	0.19	2.28	0.055	-1.045	0.090
Q2 (Hot/Dry)	30,182	10.2	0.14	6.33	0.001	-2.972	<0.001
Q3 (Hot/Wet)	30,182	9.8	0.04	0.59	0.637	-0.275	0.697
Q4 (Cool)	30,182	10.1	0.08	1.90	0.099	-0.895	0.105

Notes: Quarterly all-cause mortality estimates for the medium canopy tercile (year-2000 canopy cover 4.4–21.8% of locality area), locality-years 2002–2023. All specifications include locality and year fixed effects and quarterly population exposure. N counts outcome-specific PPML-retained locality-years used in both stages. Naive reports PPML associations without instrumenting canopy loss; CF effect reports two-step Poisson control-function IV estimates using the insect-outbreak instrument. Both columns express percentage changes in mortality at the quarter-specific first-stage shift in annual canopy loss, measured in percentage points of locality area. F is the effective first-stage F-statistic; λ is the coefficient on the first-stage residual, and p(endog) tests $\lambda = 0$. The p-value column reports CF-effect p-values from a percentile bootstrap with 1,999 locality block-resamples that re-estimate both stages; p(endog) is analytical, with standard errors clustered by locality.

Table A4: The Seasonal Thermal Environment

	Q1	Q2	Q3	Q4	Q3 – Q2	Same sign among non-ties (%)
<i>Panel A. Ambient heat</i>						
Air temperature, 24-hour mean (°C)	17.5	21.2	20.0	17.6	-1.2	87.2
Air temperature, daily max (°C)	24.7	27.7	25.0	23.5	-2.7	94.4
UTCI, daily mean (°C)	16.5	21.6	21.8	17.6	0.1	35.0
UTCI, daily max (°C)	28.3	31.6	30.4	28.3	-1.2	82.6
Days, $28 \leq T_{\text{avg}} < 32^\circ\text{C}$	1.5	10.0	8.5	1.3	-1.5	67.3
Days, $T_{\text{avg}} \geq 32^\circ\text{C}$	0.0	0.9	0.5	0.0	-0.3	73.3
<i>Panel B. The composition of heat</i>						
Relative humidity (%)	54.9	59.3	77.4	67.8	18.1	99.3
Mean radiant temp., daily max (°C)	49.5	50.9	49.1	48.4	-1.7	88.0
Surface solar radiation (MJ/m ²)	1846	2175	1874	1609	-302	99.5
Cloud frequency	0.27	0.36	0.59	0.36	0.23	98.0
Observations	31,165	31,165	31,165	31,165	62,330	

Notes: Q1–Q4 report unweighted locality-year-quarter means for 1,355 medium-cover localities (year-2000 canopy cover of 4.4–21.8% of locality area), with all ten measures observed every quarter in 2001–2023. Q3–Q2 reports coefficients on a Q3 indicator from regressions using Q2 and Q3 observations, with locality and year fixed effects; all have two-sided analytical $p < 0.005$, with standard errors clustered by locality. “Same sign” reports the percentage of localities whose 2001–2023 mean Q3–Q2 difference matches the aggregate sign, excluding zero differences. Zero differences occur in 59.9% and 82.6% of localities in the two temperature-day rows, respectively. Air temperature is ERA5-Land 2-m temperature: the 24-hour mean averages hourly values, and daily maxima are averaged within quarters. Temperature-day bins use $T_{\text{avg}} = (T_{\text{max}} + T_{\text{min}})/2$, with cut points from [Cohen and Dechezleprêtre \(2022\)](#). Relative humidity is derived daily from ERA5-Land air and dewpoint temperatures before quarterly averaging. Solar radiation is ERA5-Land total downward shortwave energy (direct and diffuse), summed to MJ/m² per quarter. UTCI (Universal Thermal Climate Index) combines air temperature, mean radiant temperature, humidity, and wind. UTCI and mean radiant temperature come from ERA5-HEAT; rows average the indicated daily statistic within each quarter ([Di Napoli et al., 2021](#)). MODIS cloud frequency is the quarterly share of daily 1-km observations classified as cloudy or mixed. All measures are sampled at locality centroids and describe conditions at the source products’ spatial resolution rather than tree-scale microclimates.

Table A5: Estimates Under the Broader Instrument-Relevant Sample

Outcome	Medium tercile		Top-two terciles	
	N	Effect (%)	N	Effect (%)
<i>Panel A: Q2 all-cause mortality, Poisson control function</i>				
Effect at first-stage shift	30,182	6.33 (0.001)	60,432	4.46 (0.001)
<i>Panel B: All-cause mortality, reduced form by season</i>				
Annual	30,226	2.58 (0.016)	60,542	1.63 (0.054)
Q2 (Hot/Dry)	30,182	6.15 (<0.001)	60,432	4.46 (<0.001)
non-Q2	30,226	1.45 (0.175)	60,542	0.74 (0.362)
Q3 (Hot/Wet)	30,182	0.53 (0.703)	60,476	-0.52 (0.597)
Q2 – Q3 contrast (pp)	60,364	5.62 (<0.001)	120,908	4.98 (<0.001)
<i>Panel C: Cause of death, Q2</i>				
Cardiovascular	29,918	7.16 (0.003)	59,948	4.87 (0.007)
Respiratory	29,258	3.22 (0.365)	58,562	1.19 (0.654)
External causes	29,280	2.58 (0.631)	58,848	3.28 (0.371)
<i>Panel D: Age group, Q2</i>				
Age 0–14	28,312	3.98 (0.210)	57,241	1.04 (0.653)
Age 15–64	30,072	4.35 (0.022)	60,256	3.01 (0.049)
Age 65+	30,089	5.91 (0.014)	60,312	4.73 (0.003)
<i>Panel E: Insurance status, Q2</i>				
Informal	29,808	8.74 (0.044)	59,882	5.13 (0.085)
Formal	27,674	2.42 (0.434)	54,910	1.39 (0.528)

Notes: Samples are the medium canopy tercile (year-2000 canopy cover of 4.4–21.8% of locality area) and the medium and high terciles pooled. The sample period is 2002–2023. Panel A reports two-step Poisson control-function IV effects of measured annual canopy loss on Q2 all-cause mortality at each sample’s first-stage shift. Panels B–E report reduced-form PPML effects of the insect-outbreak instrument. Effects are percentage changes in mortality; the Q2 – Q3 contrast is in percentage points. The contrast uses a stacked PPML regression with quarter-specific locality and year fixed effects and instrument coefficients. N counts locality-years, except in the Q2 – Q3 contrast, where it counts stacked locality-year-quarter observations. All specifications include locality and year fixed effects, with population exposure matched to the outcome’s period and age group. Q2 covers April–June; non-Q2 pools Q1, Q3, and Q4 with exposure scaled by 9/12. Insurance outcomes cover ages 15–64: informal denotes uninsured or Seguro Popular; formal comprises IMSS, ISSSTE, military, PEMEX, and private coverage. External causes include accidents and violence. N varies with outcome-specific PPML separation. P-values are in parentheses beneath each estimate. Panel A p-values use a percentile bootstrap with 1,999 locality block-resamples that re-estimate both stages; reduced-form and contrast p-values are analytical, with standard errors clustered by locality.

Table A6: Within-Q2 Reduced Forms by Solar-Radiation and Cloud-Frequency Anomalies

	Solar (MJ/m ²)	Cloud frequency	Daily-max (°C)	%Effect	<i>p</i>
<i>Panel A. Solar-radiation anomaly terciles</i>					
Low solar	2079	0.43	26.8	1.88	0.352
Typical solar	2190	0.36	28.3	12.09	0.013
High solar	2262	0.30	28.2	5.30	0.006
Low solar vs. other two terciles (difference <i>p</i>)					0.066
<i>Panel B. Cloud-frequency anomaly terciles</i>					
Cloudiest	2103	0.45	27.2	2.31	0.149
Normal	2188	0.36	28.0	12.98	0.011
Clearest	2240	0.27	28.2	4.66	0.026
Cloudiest vs. other two terciles (difference <i>p</i>)					0.067
Observations	29,544				
Localities	1,343				

Notes: Reduced-form PPML estimates of the percentage effect of the insect-outbreak instrument on Q2 (April–June) all-cause mortality in the medium canopy tercile (year-2000 canopy cover 4.4–21.8% of locality area), 2002–2023. Regressions include locality and year fixed effects and quarterly population exposure. Using locality-years with complete climate coverage, both panels classify the same Q2 locality-years into pooled terciles of deviations from each locality’s sample-period Q2 mean: solar-radiation deviations in Panel A and cloud-frequency deviations in Panel B. The classifications agree for 60.1% of these locality-years: an observation falls in both the low-solar and cloudiest groups, both the typical-solar and normal-cloud groups, or both the high-solar and clearest groups. The remaining locality-years fall in different groups under the two classifications. Climate columns report bin means: Solar is ERA5-Land quarterly surface solar radiation (MJ/m²); Cloud is the quarterly share of daily 1-km MODIS observations at the locality centroid classified as cloudy or mixed; Daily-max is the ERA5-Land quarterly mean of daily maximum temperature (°C). Each panel’s effects come from one regression interacting the instrument with tercile indicators; observation counts refer to the PPML-retained sample. Row p-values test each effect against zero. Difference p-values come from separate regressions comparing the low-solar tercile (Panel A) or cloudiest tercile (Panel B) with the other two terciles pooled. All p-values are analytical, with standard errors clustered by locality.

A.4 Who Bears the Cost

Table A7 reports the Q2 and non-Q2 estimates underlying Figure 4.

Table A8 reports the annual occupation and insurance estimates discussed in Section 4.3. The occupation data provide a limited measure of workplace exposure. Roughly half of working-age deaths in the medium-cover sample cannot be assigned to either the outdoor or indoor occupation group. Reported occupations must also be harmonized across classification systems that change over the sample, requiring broad categories that only imperfectly capture exposure to sun and shade. Quarterly death counts within these groups are sparse, leading to substantial sample loss through separation in the Poisson models. We therefore report occupation estimates only for annual mortality. Given the incomplete classification and coarse measure of exposure, we interpret the occupation comparison as suggestive evidence about occupational differences in the mortality response.

Table A9 documents how existing canopy is distributed across localities with different levels of social deprivation.

Table A7: Mortality Effects by Cause and Population: Q2 versus Non-Q2

Specification	N	F	Naive (%)	CF effect (%)	p-value	λ	p(endog)
<i>Panel A: Cause of Death, Q2 (Apr–Jun)</i>							
CV Q2	29,918	9.3	0.14	7.34	0.001	-3.597	0.004
Resp Q2	29,258	6.3	0.14	3.42	0.422	-2.471	0.365
External Q2	29,280	8.1	0.10	2.67	0.630	-1.617	0.631
<i>Panel B: Cause of Death, non-Q2</i>							
CV non-Q2	30,182	10.1	0.17	0.45	0.803	-0.141	0.889
Resp non-Q2	29,852	9.5	0.02	2.16	0.259	-1.081	0.261
External non-Q2	30,006	8.8	0.13	1.19	0.791	-0.657	0.814
<i>Panel C: Age Group, Q2 (Apr–Jun)</i>							
Age 0-14 Q2	28,312	7.5	-0.00	3.98	0.186	-2.505	0.210
Age 15-64 Q2	30,072	9.4	0.11	4.51	0.024	-2.602	0.022
Age 65+ Q2	30,089	9.1	0.13	6.07	0.002	-3.071	0.015
<i>Panel D: Age Group, non-Q2</i>							
Age 0-14 non-Q2	29,785	9.7	-0.09	1.06	0.608	-0.574	0.537
Age 15-64 non-Q2	30,204	10.1	0.08	2.51	0.127	-1.188	0.159
Age 65+ non-Q2	30,177	10.2	0.13	-0.68	0.521	0.411	0.429
<i>Panel E: Insurance Status, Q2 (Apr–Jun)</i>							
Informal Q2	29,808	8.7	0.01	8.75	0.035	-5.245	0.043
Formal Q2	27,674	7.4	0.16	2.64	0.436	-1.473	0.428
<i>Panel F: Insurance Status, non-Q2</i>							
Informal non-Q2	30,138	9.3	0.01	5.28	0.061	-3.127	0.065
Formal non-Q2	29,346	8.8	0.09	0.46	0.801	-0.187	0.831

Notes: Mortality estimates by season and subgroup for the medium canopy tercile (year-2000 canopy cover 4.4–21.8% of locality area), locality-years 2002–2023. Q2 covers April–June; non-Q2 pools Q1, Q3, and Q4. All specifications include locality and year fixed effects, with population exposure matched to the outcome’s period and age group. N counts outcome-specific PPML-retained locality-years used in both stages. Naive reports PPML associations without instrumenting canopy loss; CF effect reports two-step Poisson control-function IV estimates using the insect-outbreak instrument. Both columns express percentage changes in mortality at the outcome-specific first-stage shift in annual canopy loss, measured in percentage points of locality area. F is the effective first-stage F-statistic; λ is the coefficient on the first-stage residual, and p(endog) tests $\lambda = 0$. CV and Resp denote cardiovascular and respiratory deaths; external causes include accidents and violence. Insurance outcomes cover ages 15–64: Informal denotes uninsured or Seguro Popular; Formal comprises IMSS, ISSSTE, military, PEMEX, and private coverage. The p-value column tests each CF effect against zero using a percentile bootstrap with 1,999 locality block-resamples that re-estimate both stages; p(endog) is analytical, with standard errors clustered by locality.

Table A8: Annual Mortality Effects by Occupation and Insurance Status

Specification	N	F	Naive (%)	CF effect (%)	p-value	λ	p(endog)
<i>Panel A: Occupation</i>							
Outdoor	29,962	9.1	0.11	7.28	0.010	-4.253	0.007
Indoor	29,940	9.4	0.25	4.26	0.189	-1.995	0.198
Unclassified	30,138	9.5	0.00	0.41	0.840	-0.211	0.901
<i>Panel B: Insurance Status (ages 15–64)</i>							
Informal	30,160	9.3	0.01	6.12	0.042	-3.615	0.051
Formal	29,588	8.9	0.12	0.99	0.633	-0.443	0.648

Notes: Annual mortality estimates for the medium canopy tercile (year-2000 canopy cover 4.4–21.8% of locality area), locality-years 2002–2023. All specifications include locality and year fixed effects and use annual population exposure for ages 15–64. N counts outcome-specific PPML-retained locality-years used in both stages. Naive reports PPML associations without instrumenting canopy loss; CF effect reports two-step Poisson control-function IV estimates using the insect-outbreak instrument. Both columns express percentage changes in mortality at the outcome-specific first-stage shift in annual canopy loss, measured in percentage points of locality area. F is the effective first-stage F-statistic; λ is the coefficient on the first-stage residual, and p(endog) tests $\lambda = 0$. Outdoor combines primary-sector and sales/services workers; Indoor combines technical/manual and professional workers; Unclassified comprises working-age deaths not assigned to either group. Informal denotes uninsured or Seguro Popular; Formal comprises IMSS, ISSSTE, military, PEMEX, and private coverage. The p-value column tests each CF effect against zero using a percentile bootstrap with 1,999 locality block-resamples that re-estimate both stages; p(endog) is analytical, with standard errors clustered by locality.

Table A9: Baseline Canopy Cover and Social Lag Across Urban Localities

<i>Panel A: Mean canopy cover by social-lag tercile</i>			
Least deprived	10.3%		
Middle	18.8%		
Most deprived	27.0%		
<i>Panel B: Change in canopy cover per standard deviation of social lag</i>			
Specification	Percentage points	p-value	N
Raw	7.53	<0.001	4,197
Built surface, population, and area controls	4.85	<0.001	4,197
Municipality fixed effects, built-surface control	1.17	0.003	2,465

Notes: Cross-sectional associations between year-2000 canopy cover and social deprivation. The matched sample contains 4,197 urban localities, including 228 with no measured baseline canopy. The mortality panel excludes these zero-canopy localities but includes 180 positive-canopy localities without Social Lag information, yielding 4,149 localities, of which 4,110 are retained in the annual mortality regressions. We retain zero-canopy localities here to describe the distribution of canopy across urban localities, including its absence. Deprivation is measured by CONEVAL's year-2000 Social Lag Index (*Indice de Rezago Social*); higher values indicate greater multidimensional deprivation. Panel A reports mean canopy cover as a percentage of locality area by tercile of this index. Panel B reports linear-regression coefficients in percentage points of canopy cover per standard deviation of the index in the matched sample. The second row controls for log population in 2010, log locality area, and the year-2000 built-surface share from the Global Human Settlement Layer. The third replaces population and area controls with municipality fixed effects, retaining the built-surface share. The fixed-effects row excludes the 1,732 localities that are the only matched locality in their municipality. P-values are analytical, using heteroskedasticity-robust standard errors in the first two rows and municipality-clustered standard errors in the third.

A.5 Reduced-Form Results

Table A10 reports the reduced-form counterparts to the quarterly all-cause estimates in Figure 3 and the Q2/non-Q2 subgroup estimates in Figure 4. These estimates regress each mortality outcome directly on the insect-outbreak instrument and therefore do not divide by the first stage.

Table A10: Reduced-Form Mortality Estimates: Medium Canopy Tercile

Specification	N	RF Effect (%)	p-value
<i>Panel A: Quarterly All-Cause Mortality</i>			
Q1 (Cool/Dry)	30,182	2.06	0.092
Q2 (Hot/Dry)	30,182	6.15	<0.001
Q3 (Hot/Wet)	30,182	0.53	0.703
Q4 (Cool)	30,182	1.80	0.104
<i>Panel B: Cause $\times$ Season</i>			
CV Q2	29,918	7.16	0.003
CV non-Q2	30,182	0.26	0.896
Resp Q2	29,258	3.22	0.365
Resp non-Q2	29,852	2.14	0.261
External Q2	29,280	2.58	0.631
External non-Q2	30,006	1.07	0.813
<i>Panel C: Age $\times$ Season</i>			
Age 0-14 Q2	28,312	3.98	0.210
Age 0-14 non-Q2	29,785	1.13	0.540
Age 15-64 Q2	30,072	4.35	0.022
Age 15-64 non-Q2	30,204	2.41	0.163
Age 65+ Q2	30,089	5.91	0.014
Age 65+ non-Q2	30,177	-0.83	0.406
<i>Panel D: Insurance $\times$ Season</i>			
Informal Q2	29,808	8.74	0.044
Informal non-Q2	30,138	5.26	0.066
Formal Q2	27,674	2.42	0.434
Formal non-Q2	29,346	0.34	0.839

Notes: Reduced-form PPML estimates regressing mortality directly on the insect-outbreak instrument, reported as percentage changes. The sample is the medium canopy tercile (year-2000 canopy cover 4.4–21.8% of locality area), locality-years 2002–2023. All specifications include locality and year fixed effects, with population exposure matched to the outcome’s period and age group. N counts outcome-specific PPML-retained locality-years. Panel A reports quarterly all-cause mortality; Panels B–D report Q2 and non-Q2 mortality by cause, age, and insurance status. Q2 covers April–June; non-Q2 pools Q1, Q3, and Q4. CV and Resp denote cardiovascular and respiratory deaths; external causes include accidents and violence. Insurance outcomes cover ages 15–64: Informal denotes uninsured or Seguro Popular; Formal comprises IMSS, ISSSTE, military, PEMEX, and private coverage. P-values are analytical, with standard errors clustered by locality.

Appendix B Robustness

B.1 Falsifications and Environmental Controls

The falsifications in Table B1 examine whether nearby disturbances or common environmental conditions could explain the Q2 all-cause mortality response. The low-cover comparison applies the insect-outbreak instrument to localities with little baseline canopy, where it produces essentially no measured canopy response. Mistletoe and fire retain the medium-cover Q2 sample and replace insect exposure with the corresponding disturbance measure. In each specification, the first-stage diagnostic and mortality reduced form use the same PPML-retained observations, locality and year fixed effects, and locality-clustered inference. Table B1 reports the mortality estimates and confidence intervals; its note supplies first-stage diagnostics.

Pesticide use is not observed. The low-cover comparison addresses pest-control responses only insofar as nearby detections trigger similar activity across cover groups. It cannot rule out responses targeted to localities with more urban host material.

Table B2 examines whether the instrument is associated with environmental conditions that could also affect mortality. It regresses each of the 16 contemporaneous or lagged environmental measures on the instrument with locality and year fixed effects. Standardized shifts divide each coefficient by the measure’s standard deviation in its estimation sample. Although 10 associations have $p < 0.05$, none exceeds 0.1 standard deviations in absolute value. The lagged drought and aerosol associations motivate including antecedent conditions; the diagnostics do not establish that environmental confounding is absent.

In Table B1, the climate-zone specification replaces year fixed effects with Köppen-Geiger climate-zone-by-year fixed effects while retaining locality fixed effects. The realized-climate specification instead adds Q2 daily-mean temperature, surface solar radiation, relative humidity, precipitation, and the Palmer Drought Severity Index (PDSI); the drought-lag specification adds one-, two-, and three-year lags of Q2 PDSI. The smoke specifications control for Q2 aerosol optical depth and nearby fire days, then add the same calendar-year lags of both measures. The joint specification combines contemporaneous climate and smoke controls.

Missing climate or aerosol observations and the availability of lagged measures change the samples. For each controlled specification, we re-estimate the uncontrolled PPML reduced form on exactly the retained observations. The paired columns in Table B1 therefore separate the effect of conditioning from changes in sample composition.

Table B1: Q2 All-Cause Robustness and Falsification Estimates

Specification	Estimate (%)	95% CI	Same-sample benchmark (%)	<i>p</i>	<i>N</i>
<i>Preferred</i>					
Preferred medium-cover estimate	6.15	[2.81, 9.60]		< 0.001	30,182
<i>Falsifications</i>					
Low-cover insect-outbreak falsification	1.34	[-1.87, 4.64]		0.418	29,720
Medium-cover mistletoe-damage falsification	0.91	[-3.07, 5.04]		0.660	30,182
Medium-cover fire-damage falsification	-1.63	[-5.18, 2.04]		0.378	30,182
<i>Confound Controls</i>					
Climate-zone × year fixed effects	5.44	[1.61, 9.41]	6.07	0.005	30,116
Realized climate controls	6.27	[3.20, 9.42]	6.64	< 0.001	29,544
Climate plus drought lags	6.76	[3.53, 10.10]	6.83	< 0.001	26,858
Smoke and nearby fire controls	6.18	[2.97, 9.48]	6.23	< 0.001	29,987
Smoke and fire lags	6.37	[3.23, 9.61]	6.39	< 0.001	27,130
Joint climate and smoke controls	6.40	[3.39, 9.50]	6.70	< 0.001	29,405
<i>Predetermined Timing</i>					
Predetermined outbreak instrument	5.00	[2.13, 7.94]	6.23	< 0.001	28,810

Notes: PPML reduced-form percentage effects on Q2 (April–June) all-cause mortality, using locality-year observations from 2002–2023. The sample is the medium canopy tercile (baseline canopy cover 4.4–21.8% of locality area), except for the low-cover falsification (<4.4%). Regressions use quarterly population exposure and locality and year fixed effects; the climate-zone specification replaces year effects with Köppen-Geiger climate-zone-by-year effects. The mistletoe and fire rows replace the insect-outbreak instrument with the corresponding disturbance indicator. Climate controls use the weather and drought measures listed in Table B2; smoke controls use aerosol optical depth and nearby fire days. Lagged specifications add one-, two-, and three-year lags of drought or both smoke measures. The predetermined instrument uses qualifying detections in the five years preceding the mortality year. The same-sample benchmark re-estimates the preferred insect-outbreak specification on the exact observations used by each alternative. *N* counts retained locality-years. First-stage *F*-statistics for the low-cover, mistletoe, and fire falsifications are 0.1, 6.1, and 0.4, respectively, estimated on their corresponding reduced-form samples. Confidence intervals and *p*-values are analytical, with standard errors clustered by locality; *p*-values test whether the reported estimate equals zero.

Table B2: Climate, Smoke, and Fire Confound Diagnostics

Environmental measure	Units	Raw shift	Shift/SD	<i>p</i>	<i>N</i>
Daily-mean temperature	°C	-0.031	-0.007	0.006	29,810
Surface solar radiation	MJ/m ² /quarter	-10.334	-0.070	< 0.001	29,810
Relative humidity	percentage points	0.540	0.047	< 0.001	29,810
Precipitation	mm/quarter	17.151	0.095	< 0.001	29,810
PDSI	index points	-0.199	-0.077	< 0.001	29,810
Aerosol optical depth	dimensionless	-3.69e-04	-0.004	0.706	30,239
Nearby fire activity	fire-pixel-days/quarter	3.022	0.037	0.007	30,448
PDSI, one-year lag	index points	-0.125	-0.049	0.037	29,810
PDSI, two-year lag	index points	-0.021	-0.008	0.717	28,455
PDSI, three-year lag	index points	0.160	0.063	0.011	27,100
Aerosol optical depth, one-year lag	dimensionless	3.26e-04	0.003	0.765	30,238
Aerosol optical depth, two-year lag	dimensionless	0.004	0.038	0.003	28,854
Aerosol optical depth, three-year lag	dimensionless	0.008	0.077	< 0.001	27,489
Nearby fire activity, one-year lag	fire-pixel-days/quarter	-0.432	-0.005	0.661	30,448
Nearby fire activity, two-year lag	fire-pixel-days/quarter	1.785	0.023	0.155	29,064
Nearby fire activity, three-year lag	fire-pixel-days/quarter	0.932	0.012	0.434	27,680

Notes: Each row reports the coefficient from a linear regression of the indicated Q2 (April–June) environmental measure on the insect-outbreak instrument, with locality and year fixed effects. The sample comprises locality-year observations from 2002–2023 in the medium canopy tercile (baseline canopy cover 4.4–21.8% of locality area). Raw shift uses the units shown; Shift/SD divides the coefficient by the measure’s standard deviation in its estimation sample. PDSI is the Palmer Drought Severity Index; lower values indicate drier conditions. Aerosol optical depth (AOD) is the quarterly mean at 550 nm, sampled at the locality centroid. Nearby fire activity sums daily MODIS fire-pixel detections within 10 km of the locality centroid over the quarter. Lags refer to the corresponding Q2 measure one, two, or three years earlier. *N* counts retained locality-years; samples vary with data and lag availability. *P*-values are two-sided analytical tests that the coefficient equals zero, using standard errors clustered by locality.

B.2 Timing of Recorded Outbreaks

B.2.1 Excluding Contemporaneous Detections

Annual recording dates do not establish whether an outbreak detection preceded Q2 mortality within the same calendar year. We therefore define the instrument using only outbreaks detected before the mortality year, removing detections recorded contemporaneously with Q2 deaths. On identical observations, this restriction lowers the Q2 all-cause reduced form from 6.23% to 5.00%, while the 95% confidence interval remains above zero (Table B1).

B.2.2 Canopy Loss Around the Recording Visit

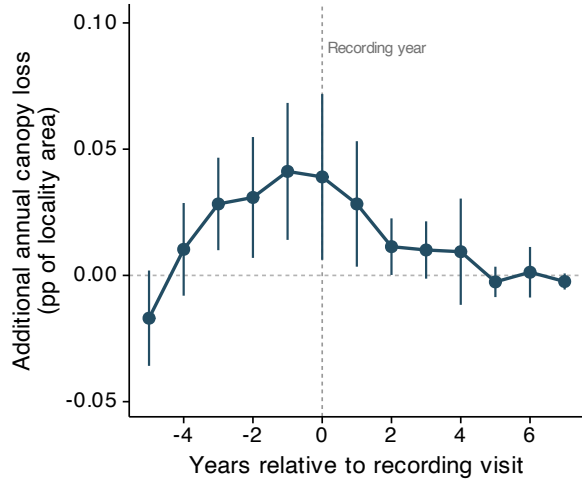
A second timing concern is that scheduled inventory visits may record outbreaks after canopy damage has begun. The preferred instrument turns on at a qualifying recording visit and remains active for five years, but its annual first-stage coefficient is a flow during that window rather than the total canopy loss associated with a recorded episode. We therefore examine canopy loss around the first recording in a new episode. We define a new recorded episode as a qualifying insect detection following five calendar years without another qualifying detection. This definition coincides with the start of a new run of the preferred five-year indicator. We restrict recording visits to 2007–2016 so that each episode has five observed years before the visit and seven after it within the 2002–2023 panel. The medium-cover sample contains 381 eligible starts, of which 368 contribute to the fixed-effects estimates.

Figure B1a reports horizon-specific regressions of Hansen canopy loss in year $t+h$ on a new episode recorded in year t , using locality-years eligible to begin a new episode and absorbing locality and calendar-year fixed effects. These regressions date canopy loss relative to the recording visit; they do not recover time relative to biological onset, which the rotating forest inventory does not observe. Figure B1b reports the corresponding cumulative specification, which sums canopy loss from year t through year $t+h$ in percentage points of locality area. To put the annual and cumulative results on the same physical scale, we evaluate these coefficients at the mean medium-cover locality area of 5.4 km². The solid series uses all recorded episode starts. The dashed series restricts to the 288 starts followed by no further qualifying detection during the next seven years, separating the response to an isolated recorded episode from extensions generated by later detections.

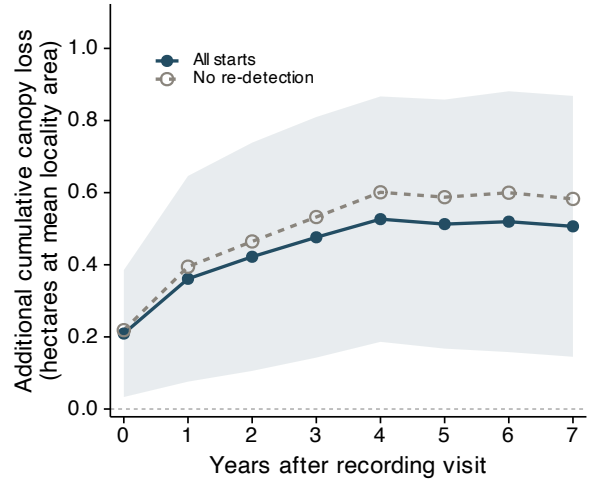
INFyS records plot conditions only during scheduled visits roughly five years apart rather than continuously observing biological onset. An outbreak may therefore begin and affect nearby canopy before the visit at which it is first recorded. Consistent with this timing, annual canopy loss is elevated during the three years before the recording visit and is largest in the year immediately preceding it. The coefficients four and five years before the visit are

not statistically significant at the 5 percent level ($p = 0.27$ and $p = 0.08$), so the detectable elevation is confined to roughly the preceding three years. The coefficients before the visit do not, however, isolate loss caused by the subsequently recorded outbreak, because antecedent conditions could contribute both to earlier canopy loss and to later outbreak detection. The “Climate plus drought lags” specification in Table B1 assesses sensitivity to antecedent drought.

After the recording visit, additional annual loss is concentrated in years zero through two and becomes small thereafter. Accordingly, cumulative loss after recording rises through the third or fourth year and then remains approximately flat. Through year four, the coefficient on cumulative loss is 0.098 percentage points of locality area (95% CI [0.035, 0.162]). Evaluated at the mean medium-cover locality area, this corresponds to 0.53 hectares (95% CI [0.19, 0.87]). This is the same area normalization used to express the annual first stage as 0.11 hectares in Section 4.1. Mechanically accumulating that annual flow over five years gives approximately 0.55 hectares, close to the cumulative event-study estimate despite the different samples and regressions. To check whether the estimated physical scale depends on using the mean locality area, we also estimate cumulative loss directly in hectares. This specification gives 0.62 hectares (95% CI [0.28, 0.96]), while measuring loss relative to each locality’s baseline canopy gives 0.98%. These alternative measures yield the same broad physical scale. Restricting to isolated recorded episodes gives 0.60 hectares under the common mean-area evaluation, indicating that the cumulative response is not driven by additional detections during follow-up. The plateau supports five years as a follow-up window after recording. The reported cumulative response excludes pre-recording loss and therefore is not total canopy loss over the biological episode.



(a) Annual loss around the recording visit



(b) Cumulative loss after the recording visit

Figure B1: Canopy Loss Around an Outbreak Recording Visit

Notes: Estimates come from separate regressions of Hansen canopy loss on a new recorded episode for each horizon, with locality and calendar-year fixed effects. Eligible locality-years have no qualifying insect detection in the previous five calendar years; a qualifying detection in the current year begins a new episode. The sample uses the medium canopy tercile and recording years 2007–2016, allowing five years before and seven years after recording within the 2002–2023 panel. Panel (a) reports annual loss in percentage points of locality area. Panel (b) reports cumulative loss from the recording year through each horizon, converted to hectares at the mean medium-cover locality area of 5.4 km². The navy estimates use 11,981 eligible locality-years, including 368 episode starts. The grey dashed series in Panel (b) excludes starts followed by another qualifying detection within seven years, using 11,898 locality-years and 288 starts. Vertical intervals in Panel (a) and the shaded band in Panel (b) are analytical 95% confidence intervals for the navy estimates, based on standard errors clustered by locality. Horizontal dashed lines mark zero; the vertical dashed line in Panel (a) marks the recording year. Recording dates do not identify biological onset, and Panel (b) excludes pre-recording loss, so it does not measure total canopy loss over the biological episode.

B.2.3 Earlier Exposure Alignment

Canopy loss is elevated in the three years before the recording visit, while the preferred instrument begins in the recording year. To assess sensitivity to this timing convention, we reconstruct the instrument to begin one, two, or three years before the visit, holding each outcome’s estimation sample fixed. The Q2 reduced form is 6.15% under this recording-year coding and 6.24%, 6.39%, and 6.59% under the three earlier alignments. The same upward pattern appears in annual mortality, where the reduced form rises from 2.58% under the recording-year coding to between 3.14% and 3.42%. The Q2 first-stage F -statistic is 10.18 under that coding and ranges from 10.67 to 11.38 under the alternatives. Each earlier alignment therefore yields a slightly larger mortality estimate without materially strengthening the first stage. These retrospective codings use a later recording to classify earlier years. They do not identify biological onset and may also capture antecedent conditions. We use them only to establish that the preferred recording-year convention does not appear to inflate the mortality estimate.

The two timing exercises change different margins: predetermined coding excludes contemporaneous detections, whereas earlier alignment allows biological damage to precede a later recording visit. Predetermined coding lowers the Q2 estimate, while earlier alignment raises it slightly. The Q2 response remains positive under both, although neither exercise identifies biological onset.

B.3 Instrument Strength and Canopy Measurement

Figure B2 examines whether the Q2 mortality response depends on the preferred instrument definition. It combines alternative spatial radii, within-plot damage thresholds, qualifying-plot shares, and carry-forward windows, holding the mortality outcome, PPML-retained sample, and fixed effects constant. The lower panel identifies the parameter choices behind each ranked estimate. First-stage diagnostics use the same observations as the reduced forms.

Table B3 examines whether the canopy first stage depends on the Hansen measurement method. It compares Hansen annual canopy loss with the annual net decline in NASA CMS continuous canopy cover on common 2002–2019 samples within each canopy tercile. CMS uses a different classifier, training source, and canopy definition. Both products share Landsat surface reflectance, so agreement across them cannot rule out measurement disturbances common to that satellite record.

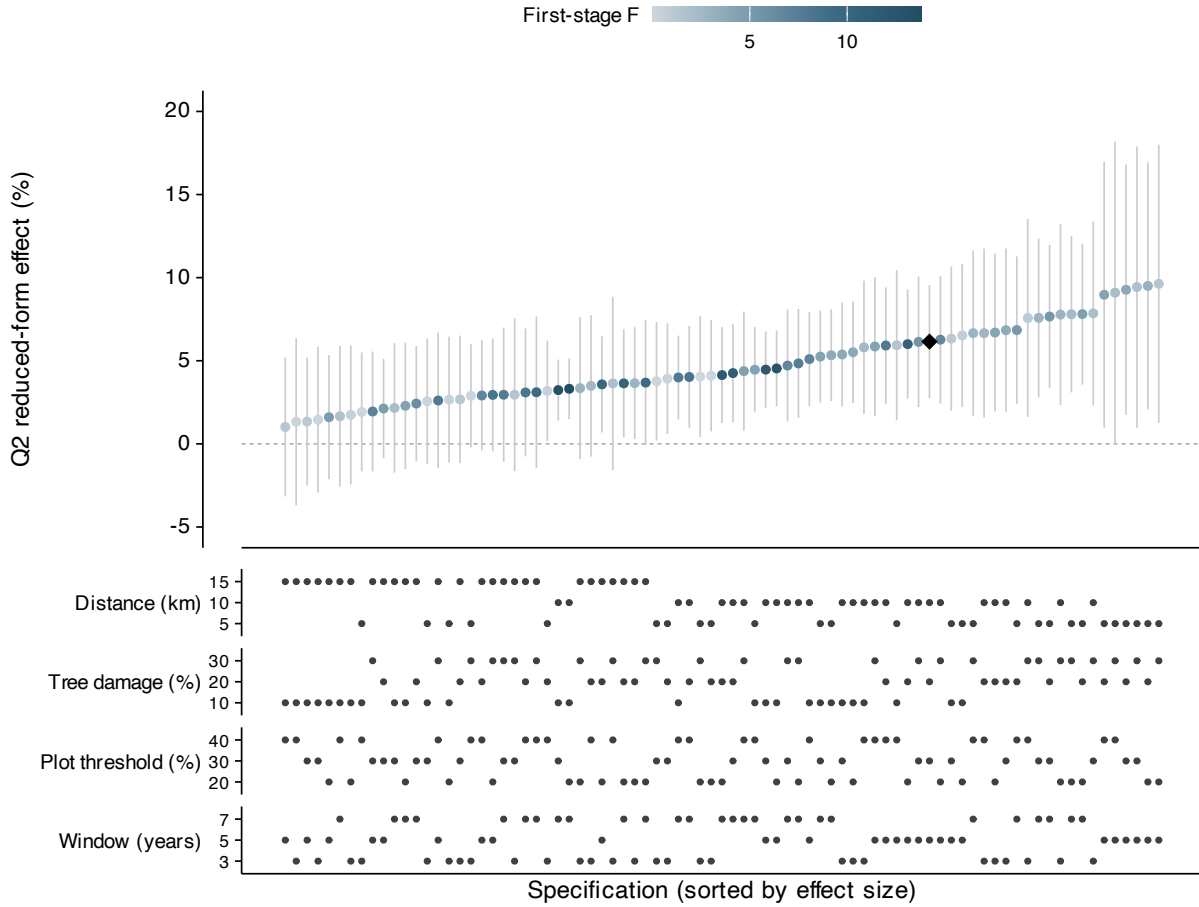


Figure B2: Instrument Specification Curve: Q2 Reduced-Form Effect Across 81 Definitions

Notes: Each point reports a PPML reduced-form percentage effect of the insect-outbreak instrument on Q2 (April–June) all-cause mortality under a different instrument definition; definitions are ranked by effect size. First stages and reduced forms use the same 30,182 PPML-retained locality-years from 2002–2023 in the medium canopy tercile. All regressions include locality and year fixed effects and cluster standard errors by locality; mortality regressions use quarterly population exposure. Vertical lines are analytical 95% confidence intervals. Point colour indicates the effective first-stage F -statistic (Montiel Olea and Pflueger, 2013); the horizontal dashed line marks zero. The lower panel identifies each definition’s distance radius from the locality boundary to CONAFOR plots, minimum percentage of sampled trees showing insect damage within a plot, minimum share of nearby plots surveyed that year meeting that damage threshold, and carry-forward window beginning with each qualifying detection. The black diamond marks the preferred values of 10 km, 20%, 30%, and 5 years, respectively.

Table B3: First-Stage Validation Using an Alternative Canopy Product

	Low (<4.4%)	Medium (4.4–21.8%)	High (>21.8%)
<i>Panel A: Hansen annual loss (% locality area)</i>			
Coefficient	0.0013 (0.0012)	0.0227 (0.0068)	0.0691 (0.0130)
First-stage F	1.1	11.2	28.4
<i>Panel B: CMS net decline (% locality area)</i>			
Coefficient	-0.0121 (0.0127)	0.0413 (0.0185)	0.0213 (0.0067)
First-stage F	0.9	5.0	10.2
Observations	24,408	24,714	24,786

Notes: Linear first-stage regressions of annual canopy-loss measures on the insect-outbreak instrument, estimated separately by baseline Hansen canopy tercile with locality and year fixed effects. Both panels use identical 2002–2019 locality-year samples within each tercile, restricted to observations retained by the annual all-cause PPML mortality regression. Panel A uses Hansen annual canopy loss; Panel B uses net decline in NASA Carbon Monitoring System (CMS) continuous tree cover ([Tran et al., 2025](#)), calculated as the previous year’s cover minus the current year’s. Coefficients are in percentage points of locality area. Standard errors clustered by locality are in parentheses; F is the cluster-robust effective first-stage F -statistic ([Montiel Olea and Pflueger, 2013](#)). Both products derive from the same Landsat surface-reflectance record.

The medium-cover Q2 effective first-stage F -statistic is 10.18, motivating a companion check using weak-instrument-robust inference. We estimate a linear 2SLS model of the Q2 mortality rate on the same 30,182 observations retained by the preferred PPML specification, with locality and year fixed effects and locality-clustered inference. We express the slope and its Anderson–Rubin confidence set on the percentage scale using the estimated first-stage shift and the sample mean mortality rate ([Anderson and Rubin, 1949](#)). The estimated effect is 7.0%, with a 95% confidence set of [1.8%, 20.7%]. The set excludes zero but permits a wide range of effect sizes. Under the monotonicity condition that outbreak exposure weakly increases measured annual canopy loss in each locality-year, the slope has an average-causal-response interpretation over the canopy-loss margins shifted by the instrument ([Angrist and Imbens, 1995](#)). The Anderson–Rubin inference applies to this linear IV model, not to the nonlinear PPML reduced form.

B.4 Inference and Multiple Testing

Table [B4](#) examines sensitivity to allowing mortality shocks to be correlated across localities and over time. The comparison uses a linear regression of the Q2 mortality rate on the instrument with locality and year fixed effects. Holding the point estimate fixed, it compares locality, municipality, and state clustering, a restricted state wild-cluster bootstrap, and Conley corrections allowing spatial and serial dependence. Its percentage effect divides the coefficient by the sample mean mortality rate. This linear sample includes observations

omitted by PPML through fixed-effect separation; the table note reconciles the samples and effect scales.

Table B4: Alternative Inference for the Q2 All-Cause Linear Reduced Form

Inference method	SE of % effect (pp)	<i>p</i> -value
<i>Linear reduced form: %-effect = 7.0%, $N = 30,446$</i>		
Locality cluster (CV1)	2.72	0.010
Municipality cluster (CV1)	2.76	0.011
State cluster (CV1, 31)	1.65	< 0.001
State wild cluster bootstrap	—	0.004
Conley spatial HAC (25–300 km $\times$ 0–7-yr windows)	2.42–3.25	0.004–0.030

Notes: Linear regression of the Q2 (April–June) all-cause mortality rate per 1,000 person-years on the insect-outbreak instrument, with locality and year fixed effects. The sample contains 30,446 locality-years from 2002–2023 in the medium baseline canopy tercile (4.4–21.8% of locality area), including 264 observations that the preferred PPML specification omits through fixed-effect separation, leaving 30,182. The percentage effect is 100 times the coefficient divided by the sample mean mortality rate. Standard errors refer to the reported percentage effect and are expressed in percentage points. This scaling differs from the multiplicative PPML effect. All rows hold the sample and point estimate fixed; *p*-values test the two-sided null of no instrument effect. Conventional cluster-robust (CV1) tests use a *t* distribution with $G - 1$ degrees of freedom, where G is the number of clusters. The state wild-cluster bootstrap imposes the null and uses 9,999 replications with Rademacher weights across 31 states (Roodman et al., 2019); it reports a *p*-value without a symmetric standard error. The Conley row reports the ranges of standard errors and *p*-values across 28 uniform-kernel specifications combining spatial cutoffs of 25, 50, 75, 100, 150, 200, and 300 km with within-locality serial-correlation windows of 0, 3, 5, and 7 years (Conley, 1999; Colella et al., 2019). Conley *p*-values use the standard normal distribution.

For the PPML specification, we also assess how unusual the observed association is relative to reassigned exposure histories. We re-estimate the reduced form after randomly reassigning complete instrument time series across localities, preserving their temporal structure. The two-sided *p*-value compares the absolute observed effect with the distribution of absolute permuted effects and equals 0.001.

To assess whether the PPML result depends on any one state, Figure B3 omits each state in turn from the medium-cover Q2 sample and re-estimates the reduced form.

Finally, the seasonal and subgroup analyses involve multiple hypotheses. Following Anderson (2008), we compute sharpened *q*-values to control the false-discovery rate within four families, using percentile-bootstrap *p*-values for the control-function mortality effects: (i) Q1–Q4 all-cause seasonality (4 tests); (ii) Q2 cardiovascular, respiratory, age-group, and informal-insurance outcomes (6 tests); (iii) Q2 external-cause and formal-insurance benchmarks (2 tests); and (iv) the non-Q2 counterparts of families (ii) and (iii) (8 tests).

Seasonal contrasts and robustness specifications revisit these outcome hypotheses. The Q2, non-Q2, and annual informal-minus-formal contrasts overlap because annual mortality

aggregates the two seasonal periods; we report their raw percentile p -values without assigning them another family. The within-Q2 beam and occupation analyses remain secondary.

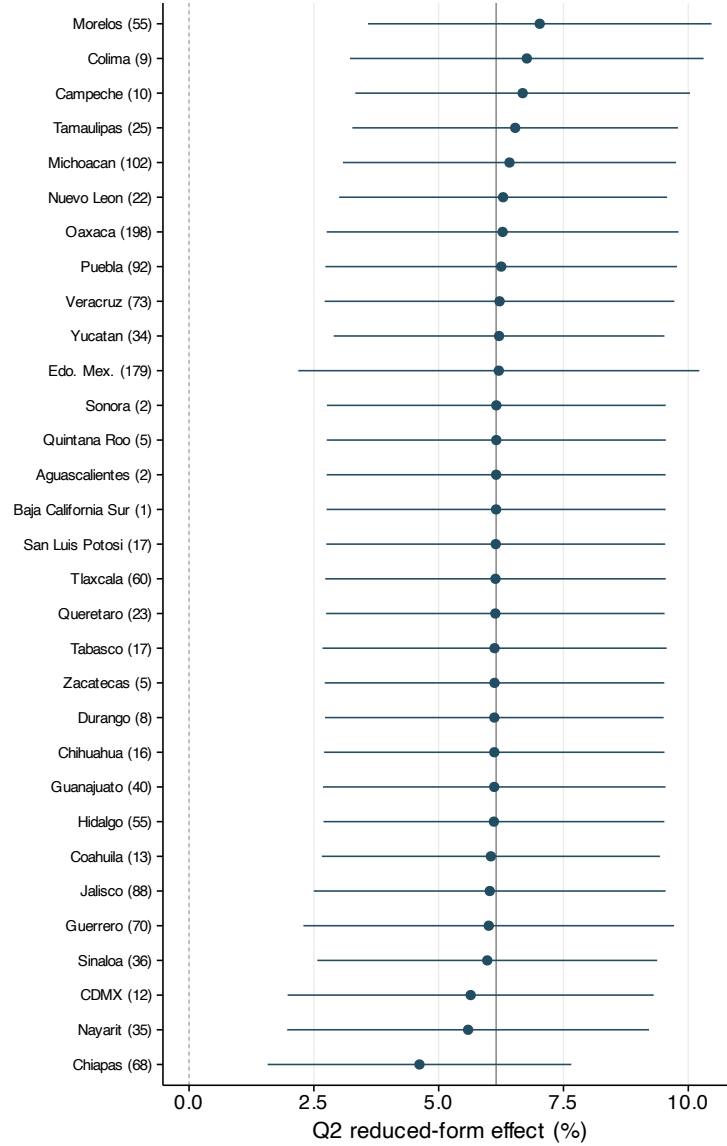


Figure B3: Leave-One-Out by State: Q2 Reduced-Form Effect

Notes: Each point reports the PPML reduced-form percentage effect of the insect-outbreak instrument on Q2 (April–June) all-cause mortality after omitting one state. The starting sample contains 30,182 locality-years across 1,372 localities in the medium canopy tercile, 2002–2023. All regressions include locality and year fixed effects, use quarterly population exposure, and cluster standard errors by locality. States are sorted by estimated effect; numbers in parentheses give the number of sample localities omitted. Horizontal lines show analytical 95% confidence intervals. The solid vertical line marks the 6.2% benchmark with all states included (Table B1); the dashed vertical line marks zero.

Appendix C The Locality-Wide Ambient Channel

This appendix asks how much of the Q2 mortality response could be explained by thermal changes measured at the locality scale. Table C1 combines the insect-outbreak instrument’s effect on each thermal measure with a mortality dose-response for that measure. We translate the resulting thermal change into a mortality contribution and compare it with the reported Q2 reduced form. The exercise uses dose-responses estimated within our sample and, for heat-day counts, an external calibration from Mexico.

The medium-cover air-temperature calculation illustrates the approach. The instrument lowers the Q2 mean of daily maximum temperature by 0.05°C , with a 95% confidence interval running from -0.083°C to -0.026°C . Although the entire interval indicates cooling, we deliberately treat its largest absolute endpoint, 0.083°C , as warming. We pair this displacement with the upper confidence limit of the in-sample temperature–mortality response: 4.89% higher mortality per degree, compared with a point estimate of 3.7%. Applying the Poisson model’s exponential transformation gives a mortality contribution of 0.40%, or 6.5% of the 6.15% Q2 reduced form. At the same upper dose-response, explaining the entire reduced form would require 1.25°C of warming, about 15 times the displacement used in the bound.

We apply this construction to each thermal measure. The in-sample dose-response comes from a Poisson regression of Q2 mortality on realized thermal conditions, using quarterly population exposure and locality and year fixed effects. It draws on within-locality weather variation over time. The instrument–temperature regression uses the same observations and fixed effects, and both regressions cluster inference by locality. We combine the largest absolute endpoint of the thermal effect’s 95% confidence interval with the upper endpoint of the dose-response’s 95% confidence interval. Under the normal approximation, these provide one-sided limits with at least 97.5% coverage each. For a nonnegative warming–mortality response, Bonferroni therefore gives at least 95% coverage for each resulting ceiling on the modeled mortality contribution without requiring independence between the estimates. We express this ceiling as a share of the canopy sample’s reported Q2 reduced form, holding that denominator at its point estimate.

For heat-day counts based on daily mean temperature, we also use an external dose-response. Cohen and Dechezleprêtre (2022) estimate a mortality response of 0.20 deaths per 100,000 people per additional hot day in Mexico, relative to a day in their $24\text{--}28^{\circ}\text{C}$ reference bin (their Figure 2). This is their largest heat coefficient; applying it to both hot-day bins, separated at 32°C , assigns the higher response to the cooler bin as well. We multiply this coefficient by the sum of the conservative instrument-induced changes in the

two counts, then compare the implied additional deaths with those attributed to the Q2 reduced form. This calculation is an external calibration: we do not propagate uncertainty in the published coefficient, and we transfer a national, station-based daily distributed-lag estimate to reanalysis exposures aggregated to the quarter.

Panel A reports the calculations across thermal measures in the medium-cover sample. The in-sample heat-day bounds are at most 1.6% of the mortality reduced form, the external heat-day calibration gives 1.4%, and the surface-temperature bound is 2.6%. Panel B repeats the air-temperature and heat-day calculations for the high-cover and pooled top-two-tercile samples. The high-cover bounds are wider: the largest is 23.9%, using the in-sample response to the count of days reaching 28°C; the air-temperature bound and external heat-day calibration are 9.4% and 10.1%, respectively. In the pooled sample, the three reported shares are at most 4.5%.

The calculation compares the instrument’s effects on realized thermal conditions and mortality directly, so it requires no rescaling by the measured canopy first stage. It therefore allows for thermal responses to canopy changes that the 30-meter Hansen product may miss, including partial thinning and small-tree loss. Its scope nevertheless depends on the thermal products’ spatial resolution. MODIS surface temperature averages one-kilometer pixels within each locality polygon, while ERA5 measures come from the coarser reanalysis cell containing the locality centroid. Neither resolves exposure at the street or courtyard where canopy stands. The exercise limits mortality contributions from thermal changes visible in these locality-scale measures; direct-beam shielding and radiation-driven cooling of nearby surfaces and air may operate below that resolution. The seasonal and cause-of-death evidence in the main text provides separate support for that localized shade interpretation.

Table C1: Bounding the Locality-Wide Thermal Channel

	Dose-response	Thermal effect $\hat{\tau}$ Estimate [95% CI]	Mortality response $\hat{\delta}$ Estimate [95% CI]	Max share of Q2 RF (%)	Change to generate RF (multiple of τ_U)
<i>Panel A. Q2 ambient measures, medium canopy tercile (Q2 all-cause RF = 6.2%)</i>					
Daily-max air temp, °C (9 km)	In-sample	−0.054 [−0.083, −0.026]	3.65%/°C [2.43, 4.89]	6.5	+1.25°C (15×)
Land-surface temp, °C (1 km)	In-sample	−0.063 [−0.130, 0.004]	0.60%/°C [−0.03, 1.23]	2.6	+4.9°C (38×)
Days, daily mean $\geq 28^\circ\text{C}$	In-sample	−0.004 [−0.228, 0.221]	0.16%/day [−0.02, 0.34]	1.2	+18 d (78×)
Days, daily mean $\geq 32^\circ\text{C}$	In-sample	+0.133 [0.068, 0.197]	0.18%/day [−0.16, 0.51]	1.6	+12 d (59×)
Heat days ($28 \leq T < 32, T \geq 32$)	External	0.55 d [†]	$\theta=0.20$	1.4	+41 d (74×)
<i>Panel B. Locality-wide thermal bounds across canopy-cover samples</i>					
	First-stage $\hat{\pi}$ (pp)	Own Q2 RF (%)	Max % (air temp)	Max % (days, daily mean $\geq 28^\circ\text{C}$)	Max % (external heat)
Medium	0.020	6.2	6.5	1.2	1.4
High	0.053	1.8	9.4	23.9	10.1
Top two terciles	0.036	4.5	3.2	4.5	2.3

Notes: Bounds on the share of the Q2 (April–June) all-cause mortality reduced form (RF) attributable to locality-scale thermal changes. Observations are locality-years, 2002–2023. Panel A uses the medium canopy tercile (baseline cover 4.4–21.8% of locality area); Panel B compares Medium, High, and their pooled sample. The mortality RF samples contain 30,182, 30,250, and 60,432 observations, respectively; air-temperature and heat-day estimates use 29,544, 29,634, and 59,178. Medium-cover land-surface temperature uses 30,177 observations. On the medium-cover air-temperature sample, the mortality RF is 6.64%, compared with the reported 6.15% used in the bound. All regressions include locality and year fixed effects; mortality regressions use PPML with quarterly population exposure. $\hat{\tau}$ is the linear reduced-form effect of the insect-outbreak instrument on the thermal measure; in-sample $\hat{\delta}$ is the percentage mortality response per degree or additional heat day. Bracketed intervals beneath estimates are analytical 95% confidence intervals based on locality-clustered standard errors; dose-response intervals are transformed from the PPML log scale and use the same units as the estimates. Define the conservative thermal displacement as $\tau_U = |\hat{\tau}| + 1.96 SE(\hat{\tau})$. Maximum in-sample shares apply the upper 95% dose-response to τ_U , using the Poisson transformation, and express the resulting contribution as a percentage of the reported RF, held fixed. Required change uses the same upper dose-response; parenthetical multiples divide that change by τ_U . Calculations use unrounded estimates. Appendix C explains the confidence construction. The external calibration applies [Cohen and Dechezleprêtre \(2022\)](#)’s coefficient of 0.20 deaths per 100,000 per additional day relative to a 24–28°C day to both daily-mean temperature bins, $28 \leq T < 32^\circ\text{C}$ and $T \geq 32^\circ\text{C}$, without propagating its sampling uncertainty. The dagger marks the sum of the two conservative thermal displacements; the external share compares implied additional deaths with deaths attributed to the reported RF, and its parenthetical multiple divides required change by that combined displacement. Daily mean temperature is $(T_{\max} + T_{\min})/2$. In Panel B, $\hat{\pi}$ is the instrument’s effect on measured annual canopy loss, in percentage points of locality area. MODIS surface temperature averages one-kilometer pixels within locality polygons; ERA5 measures are sampled at locality centroids.